Vietnam National University of HCMC

International University

School of Computer Science and Engineering

Program of MIT

Course: Advanced Computer Graphics (IT516)

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Transformation and Perspective (24/3/2026)

- Textbook: chapter 7
- Mathematics for CG: chapter 9
- Intro to CG using Java 2D & 3D: chapter 5

Transformations and Matrices

- Transformations are functions
- Matrices are function representations
- Matrices represent linear transformations
- $\{2x2 \text{ Matrices}\} \equiv \{2D \text{ Linear Transf's}\}$

Recall: computing between matrices

- Adding, subtracting between matrix A and B:
  - they must have the same degree
  - [A(m, n)] + [B(m, n)] = [C(m, n)]
- Multiple between matrix A and B:
  - the number of column of matrix A equal the number of row of matrix B:
  - [A(m, n)]x[B(n, p)] = [C(m, p)]

What is a *2D Linear Transformation*?

Recall from Linear Algebra:

$$\underline{Def}: \quad T(a\vec{x} + \vec{y}) = aT(\vec{x}) + T(\vec{y}),$$

for scalar a and vectors $\vec{x}$ and $\vec{y}$.

Look at a Diagram

$$\begin{tikzcd}
\vec{x}, \vec{y} \arrow[r, "T()"] \arrow[d, "a\times[\ ]+\ b\times[\ ]"'] & T(\vec{x}), T(\vec{y}) \arrow[d, "a\times[\ ]+\ b\times[\ ]"'] \\
a\vec{x}, b\vec{y} \arrow[r, "T()"] & \begin{aligned} & aT(\vec{x})+bT(\vec{y}) \\ & = T(a\vec{x})+T(b\vec{y}) \end{aligned}
\end{tikzcd}$$

Scale in x by 2: $S_{2_x}(v)$

- What is Scaling?
  - A scaling leads to stretching or shrinking of objects in the direction of the x-axis and the y-axis.

Scale in x, by 2, say:

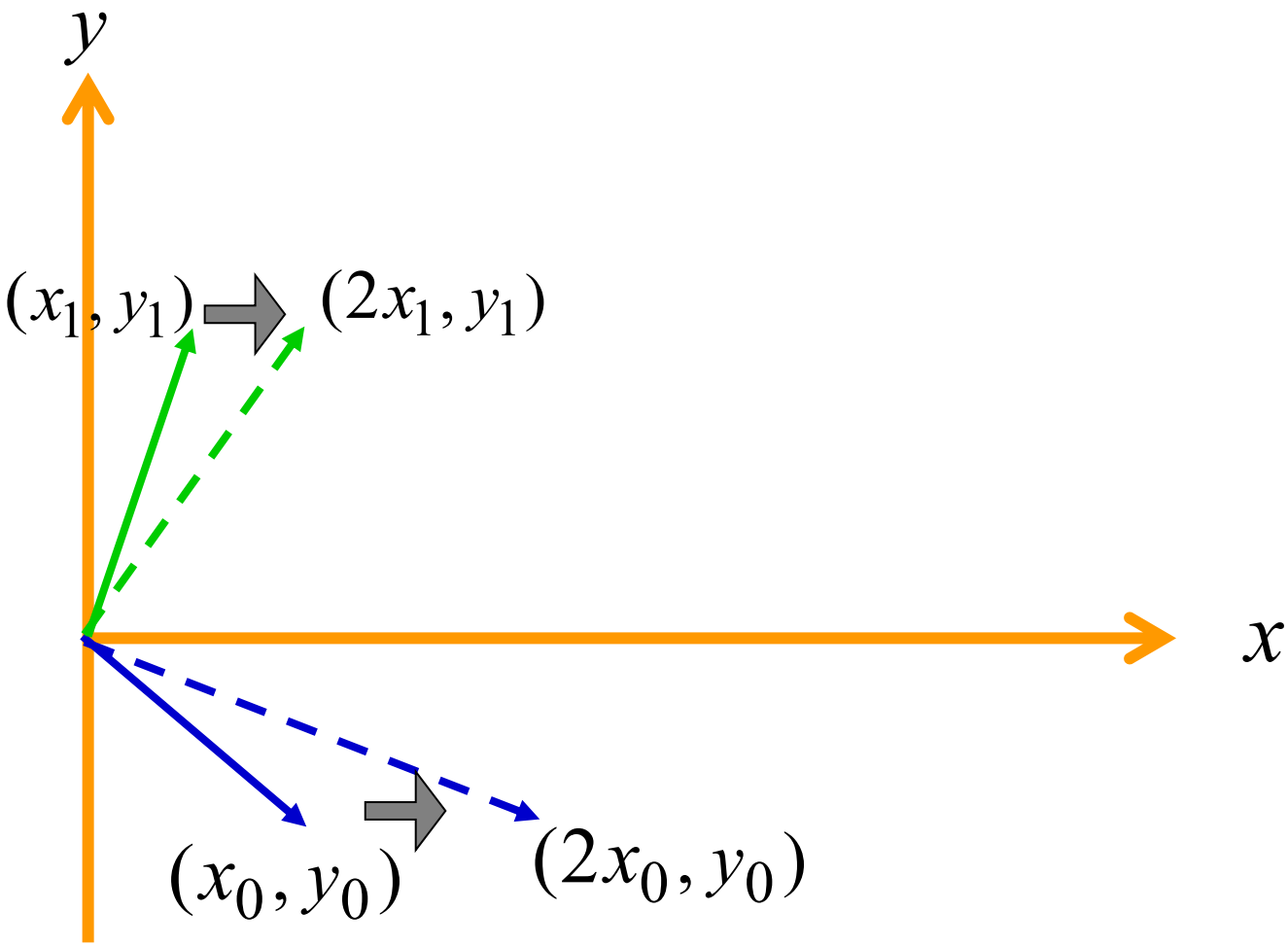
$$\begin{aligned}\left(2\left(x_0+x_1\right), y_0+y_1\right) &= \left(2x_0+2x_1, y_0+y_1\right)\\ &= \left(2x_0, y_0\right)+\left(2x_1, y_1\right)\end{aligned}$$

Example:

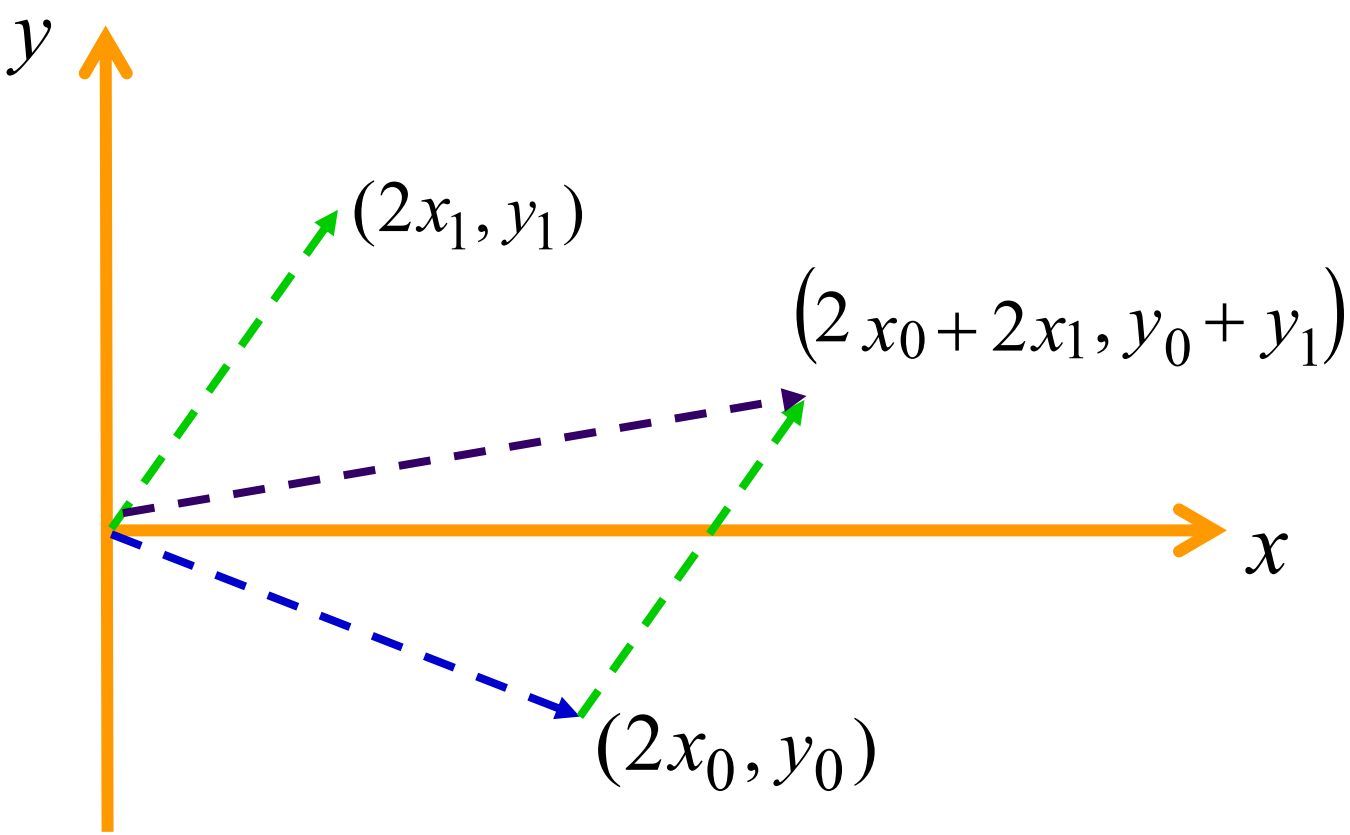
Scale *in* x *by* $S_{2_x}(v)$

What is the graphical view?

Scale in x by 2: $S_{2_x}(v)$



$$\left(2x_0 + 2x_1, y_0 + y_1\right)$$



Matrix Representation

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ y \end{bmatrix}$$

Matrix Representation of $S_{2_y}(v)$

<u>Scale *in* y</u> *by 2:* $\mathbf{S}_{2_y}(\mathbf{v})$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2y \end{bmatrix}$$

Matrix Representation $S_2(v)$

Overall *Scale by 2*: $S_2(v)$

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix}$$

Matrix Form of Same

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_0 + x_1 \\ y_0 + y_1 \end{bmatrix} = \begin{bmatrix} 2\left(x_0 + x_1\right) \\ y_0 + y_1 \end{bmatrix}$$

1 4 4 2 4 4 3

Add x and y, then scale

$$= \begin{bmatrix} 2x_0 + 2x_1 \\ y_0 + y_1 \end{bmatrix}$$

1 4 2 4 3

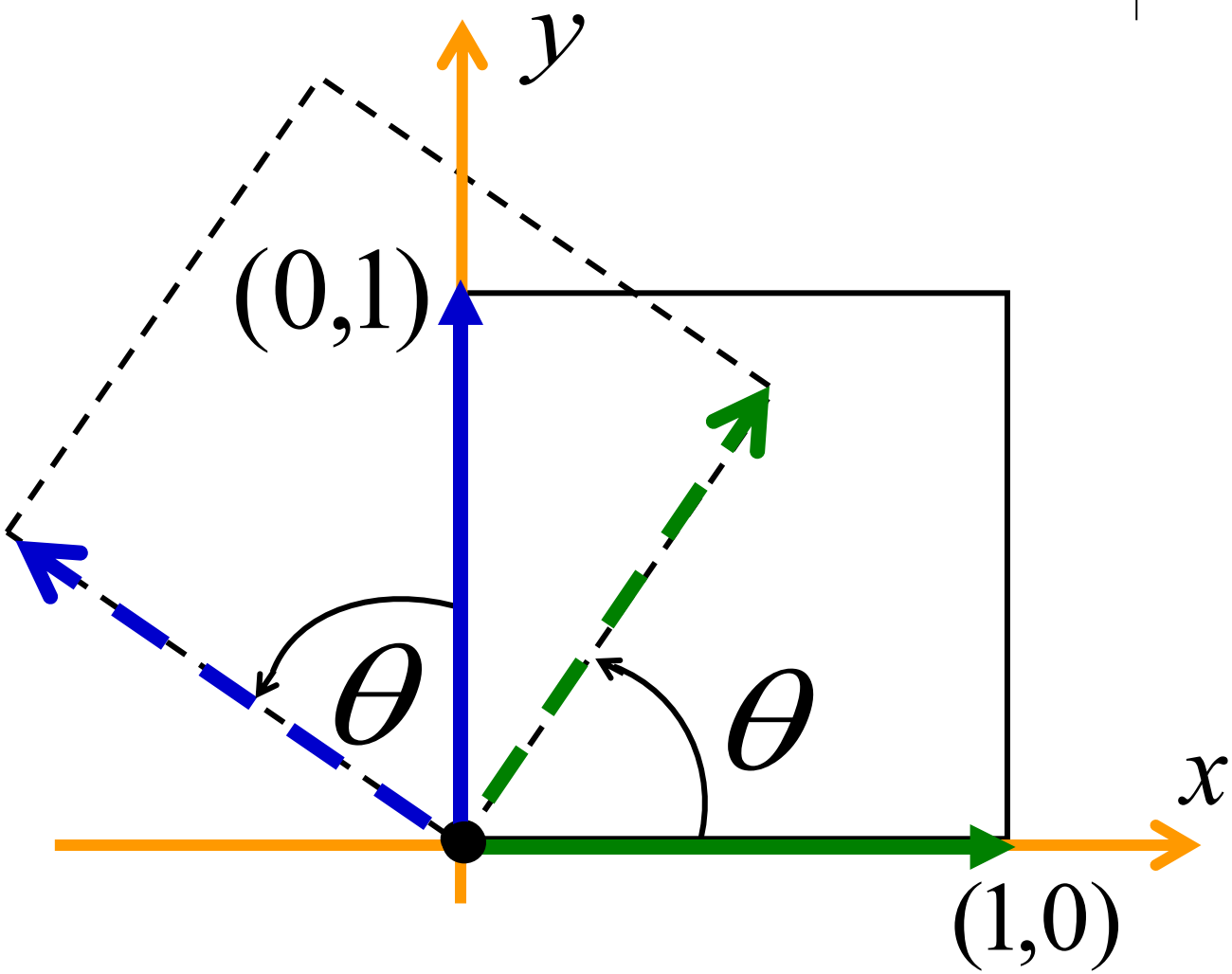
Scale x and y, then add

What about Rotation?

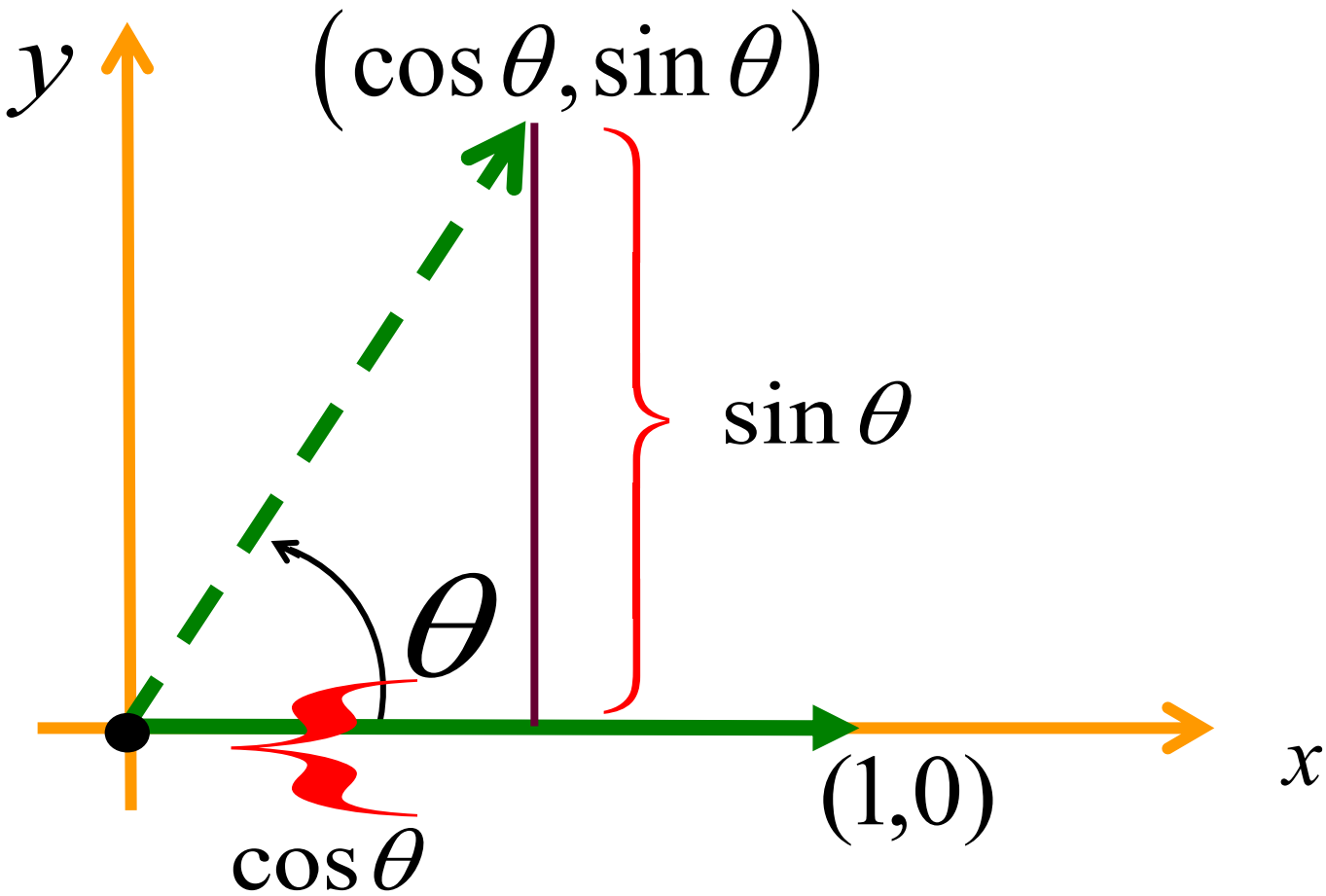
Is it linear?

- Rotations are determined by a single parameter, the rotation angle “θ”.
- Clockwise or anticlockwise around the origin of the coordinate system.

Rotate by θ



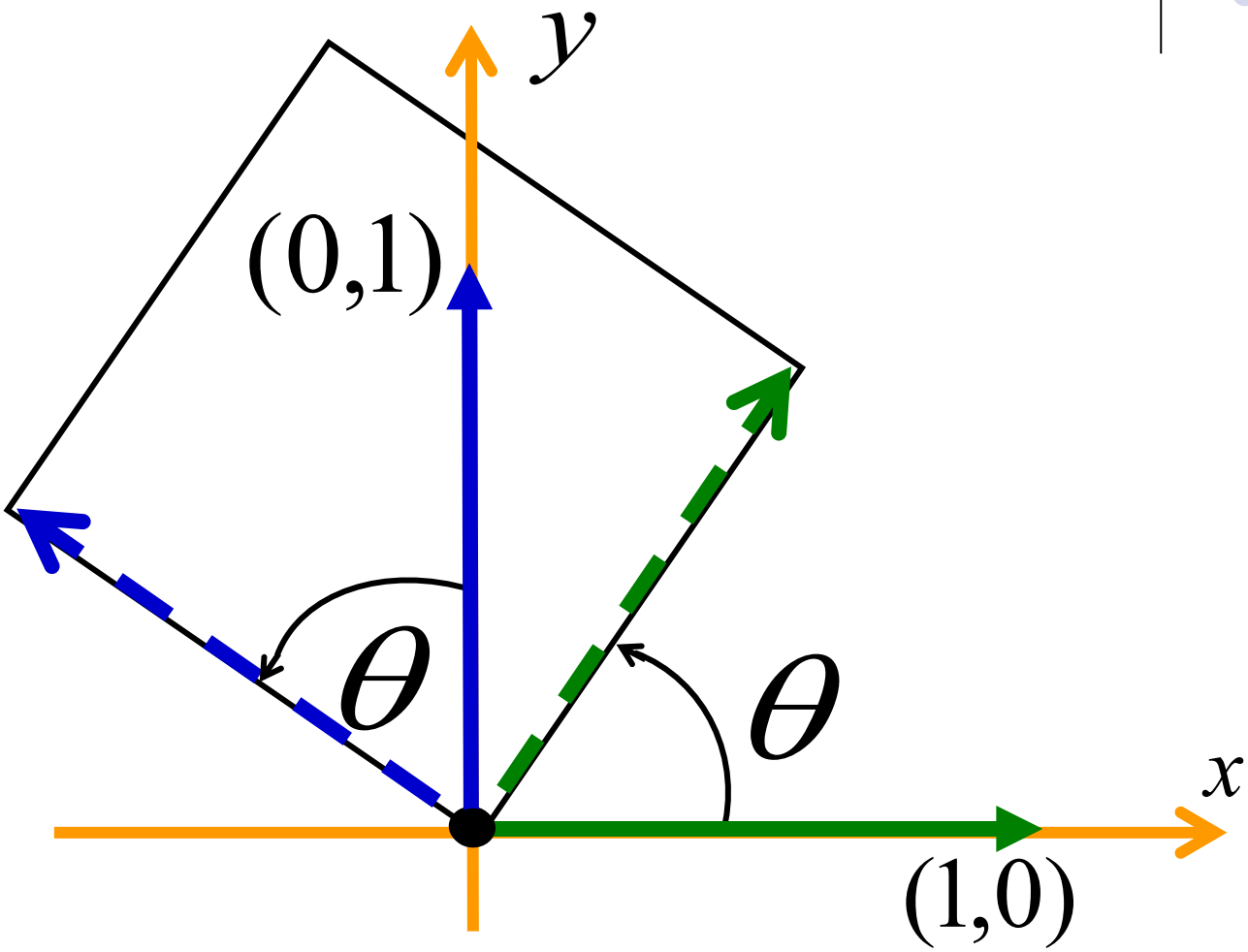
Rotate by θ : 1st *Quadrant*



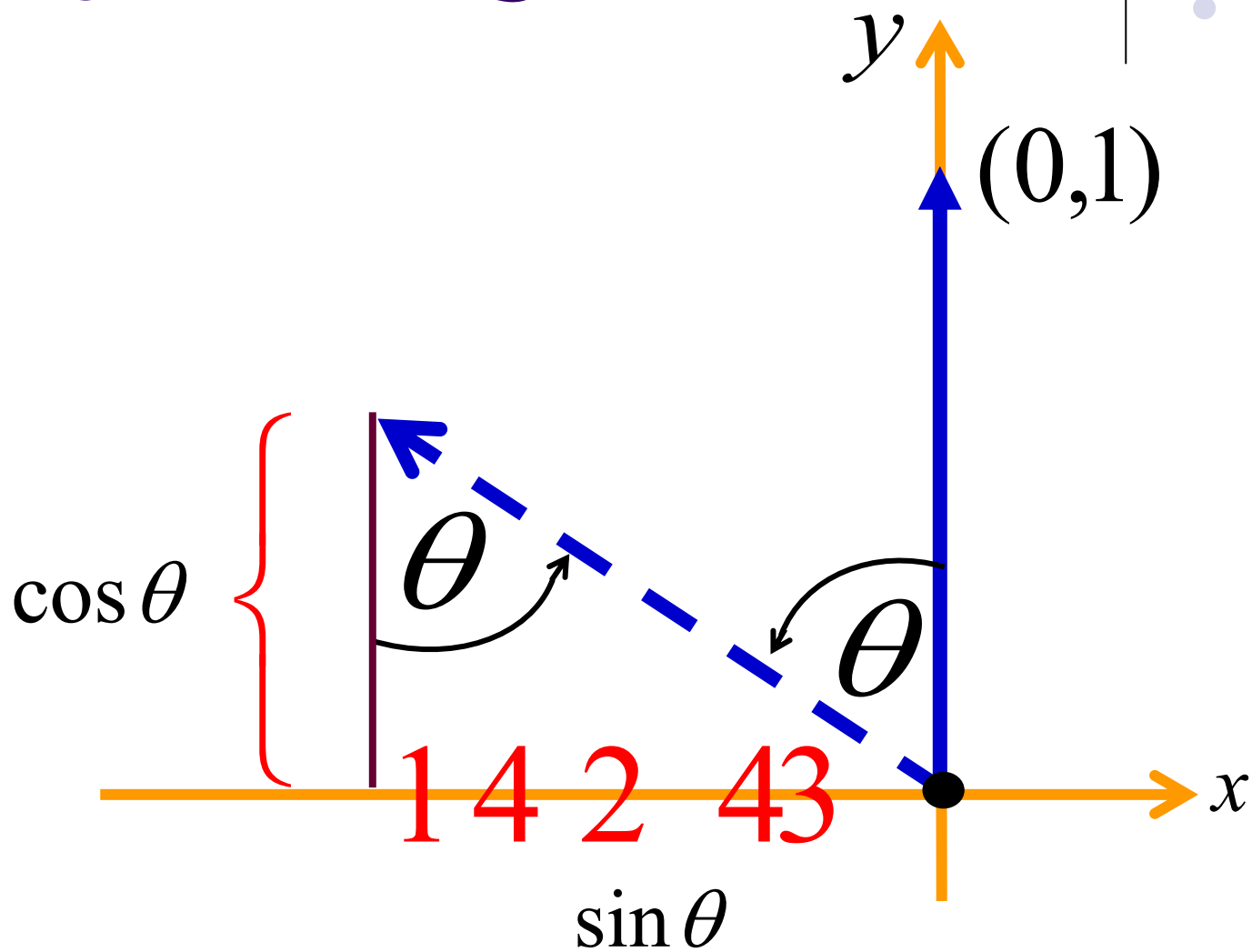
Rotate by θ: 1st *Quadrant*

$$(1,0) \Rightarrow (\cos\theta, \sin\theta)$$

Rotate by θ : 2nd *Quadrant*



Rotate by θ : 2nd *Quadrant*



Rotate by θ : 2nd *Quadrant*

$$(0,1) \Rightarrow (-\sin\theta, \cos\theta)$$

Summary of *Rotation by* θ

$$(1,0) \Rightarrow (\cos\theta, \sin\theta)$$

$$(0,1) \Rightarrow (-\sin\theta, \cos\theta)$$

Summary (Column Form)

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

Using Matrix Notation

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

(Note that *unit vectors* simply copy columns)

General *Rotation by* θ Matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

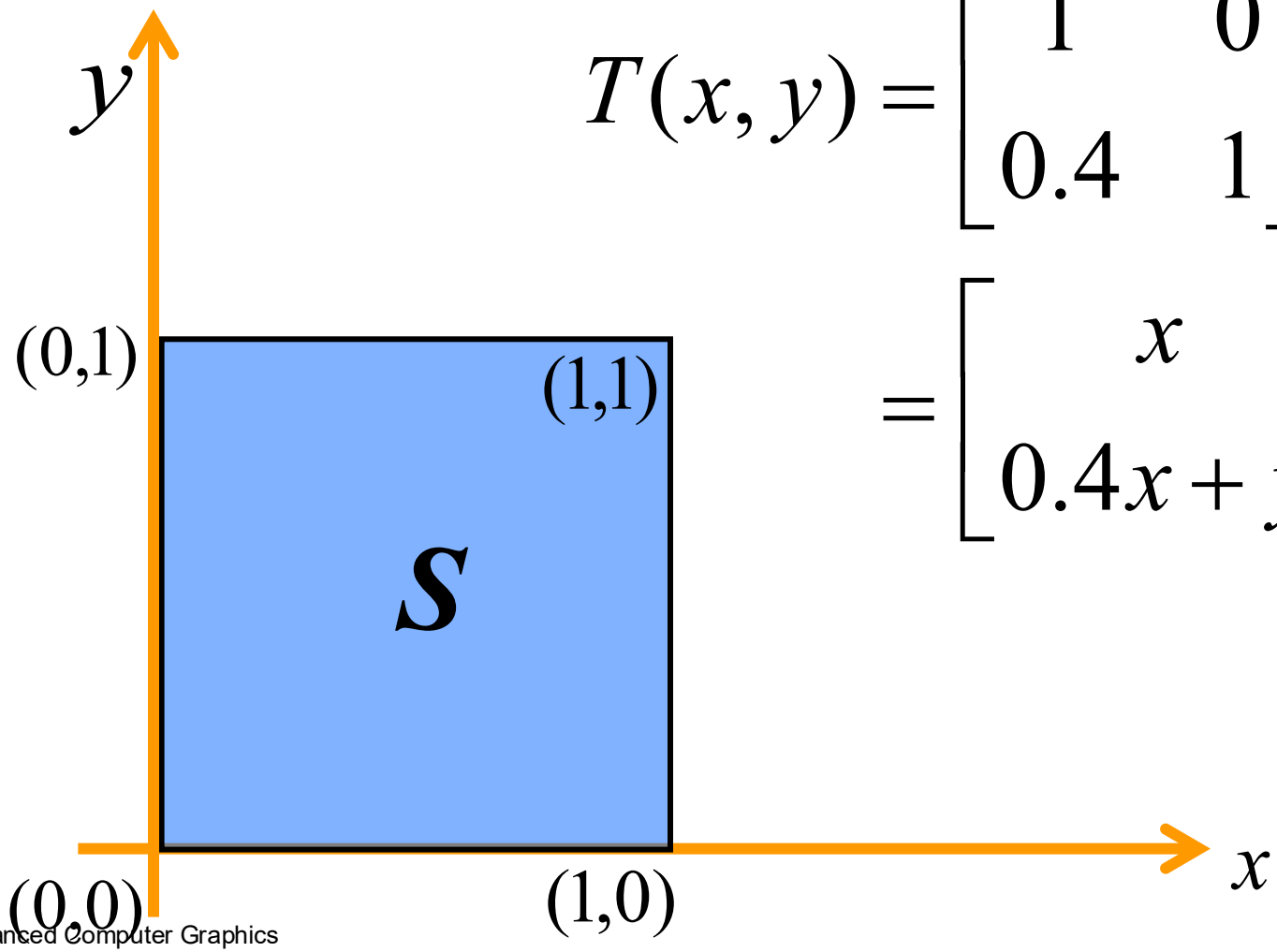
Examples of Translate and Shear

Off Diagonal Elements

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + a\,y \\ y \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ b\,x + y \end{bmatrix}$$

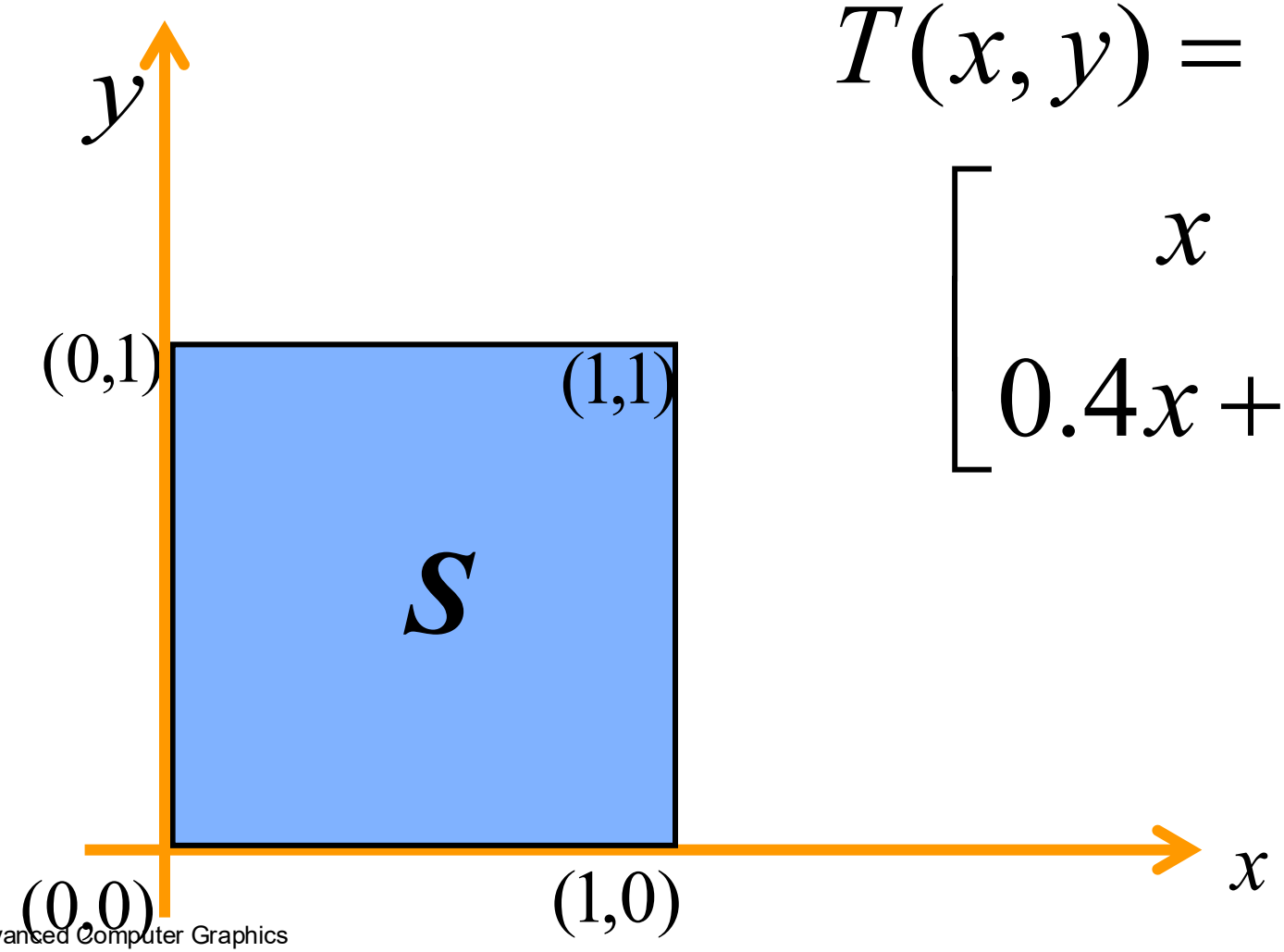
Example 1



$$T(x,y) = \begin{bmatrix} 1 & 0 \\ 0.4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

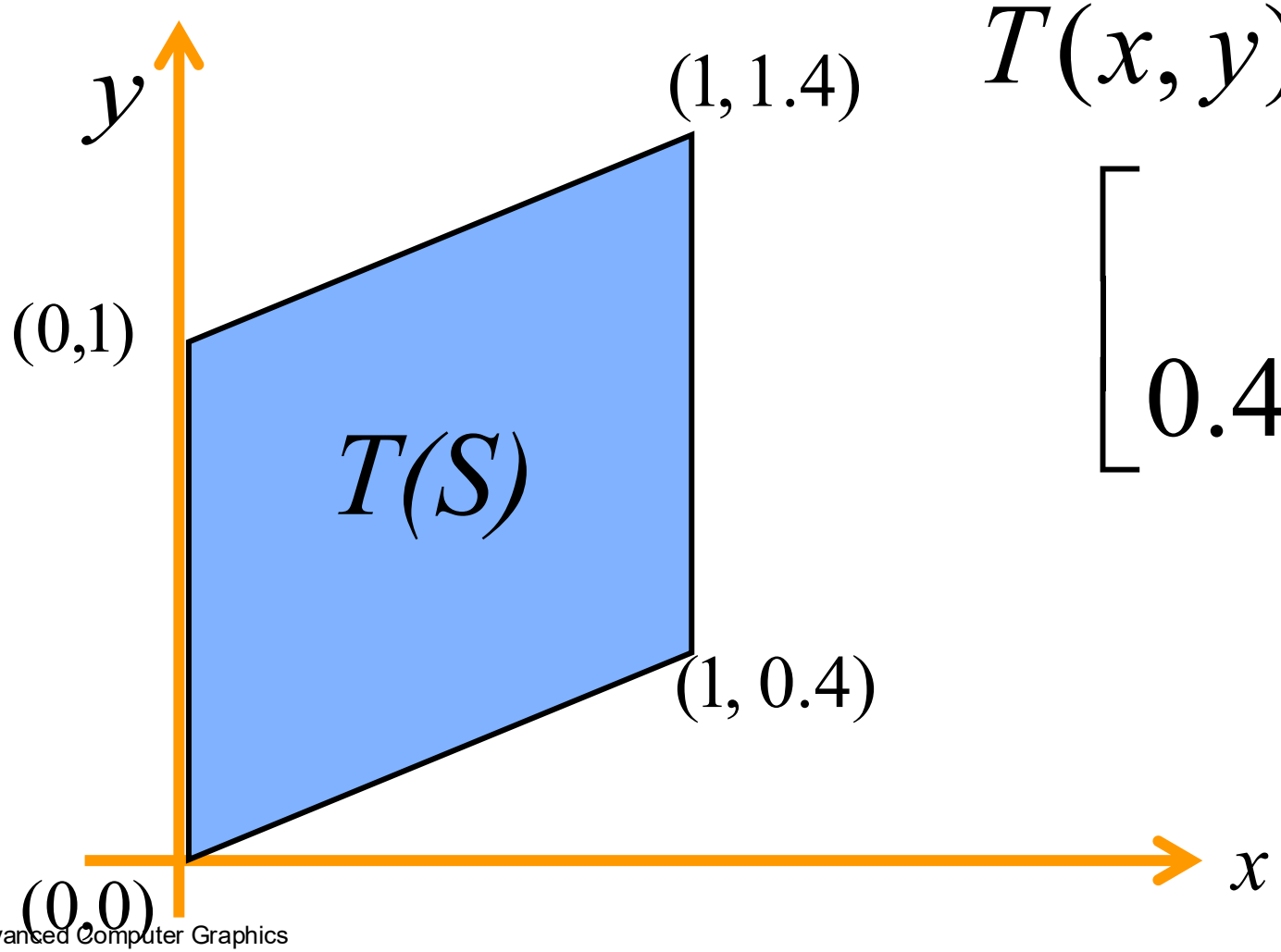
$$= \begin{bmatrix} x \\ 0.4x + y \end{bmatrix}$$

Example 1



$$T(x, y) = \begin{bmatrix} x \\ 0.4x + y \end{bmatrix}$$

Example 1

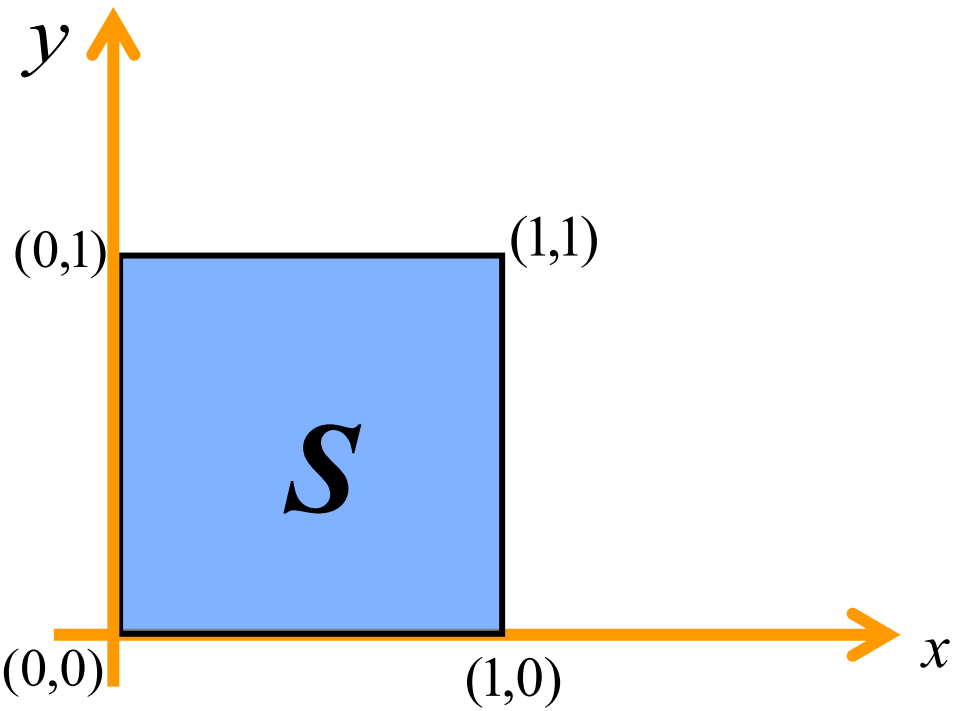


$$T(x, y) = \begin{bmatrix} x \\ 0.4x + y \end{bmatrix}$$

Example 2

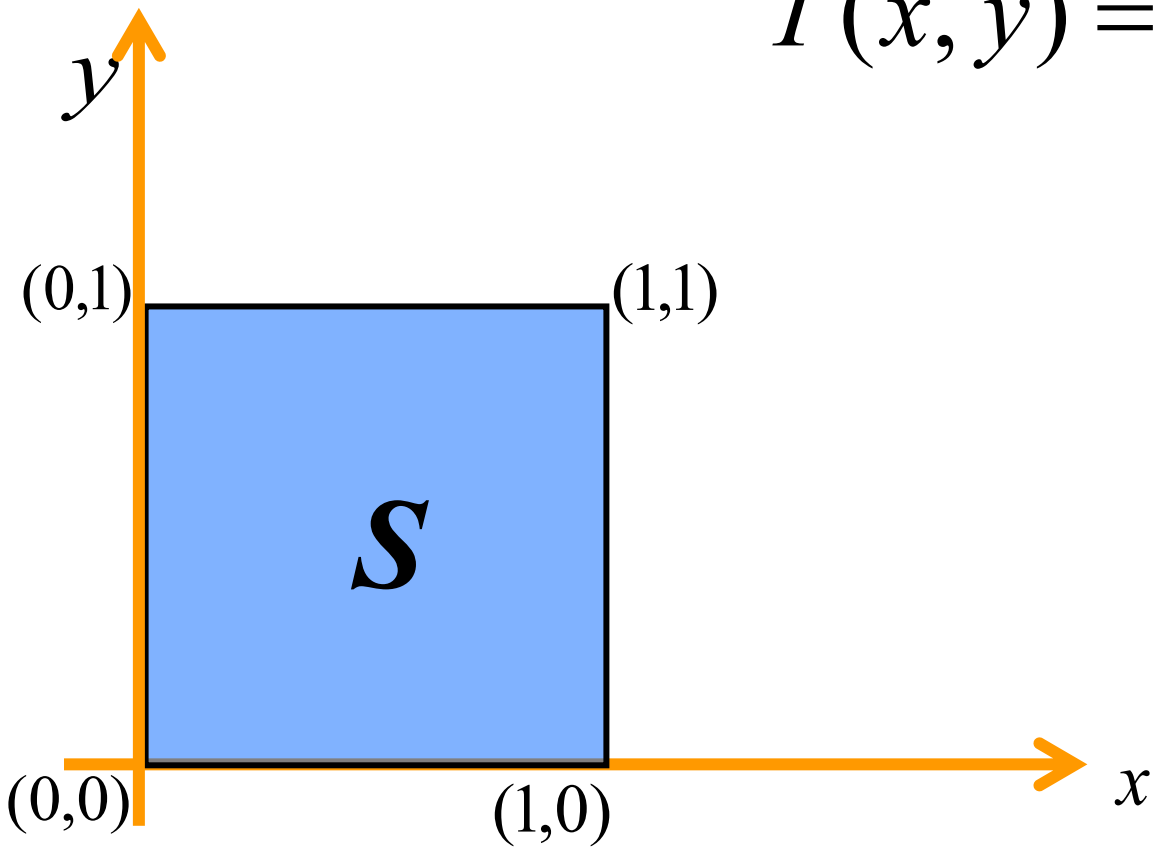
$$T(x,y) = \begin{bmatrix} 1 & 0.6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} x + 0.6y \\ y \end{bmatrix}$$



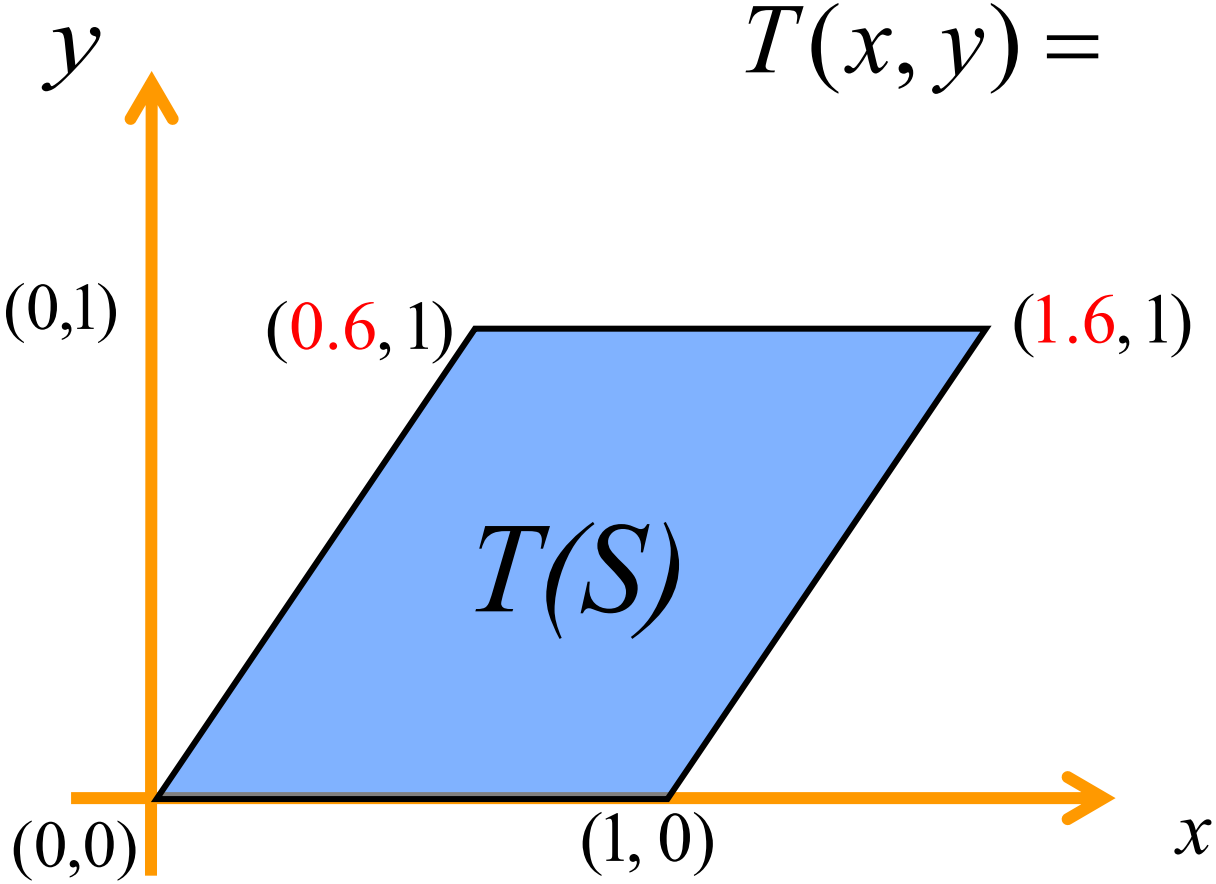
Example 2

$$T(x,y) = \begin{bmatrix} x + 0.6y \\ y \end{bmatrix}$$



Example 2

$$T(x, y) = \begin{bmatrix} x + 0.6y \\ y \end{bmatrix}$$



Summary

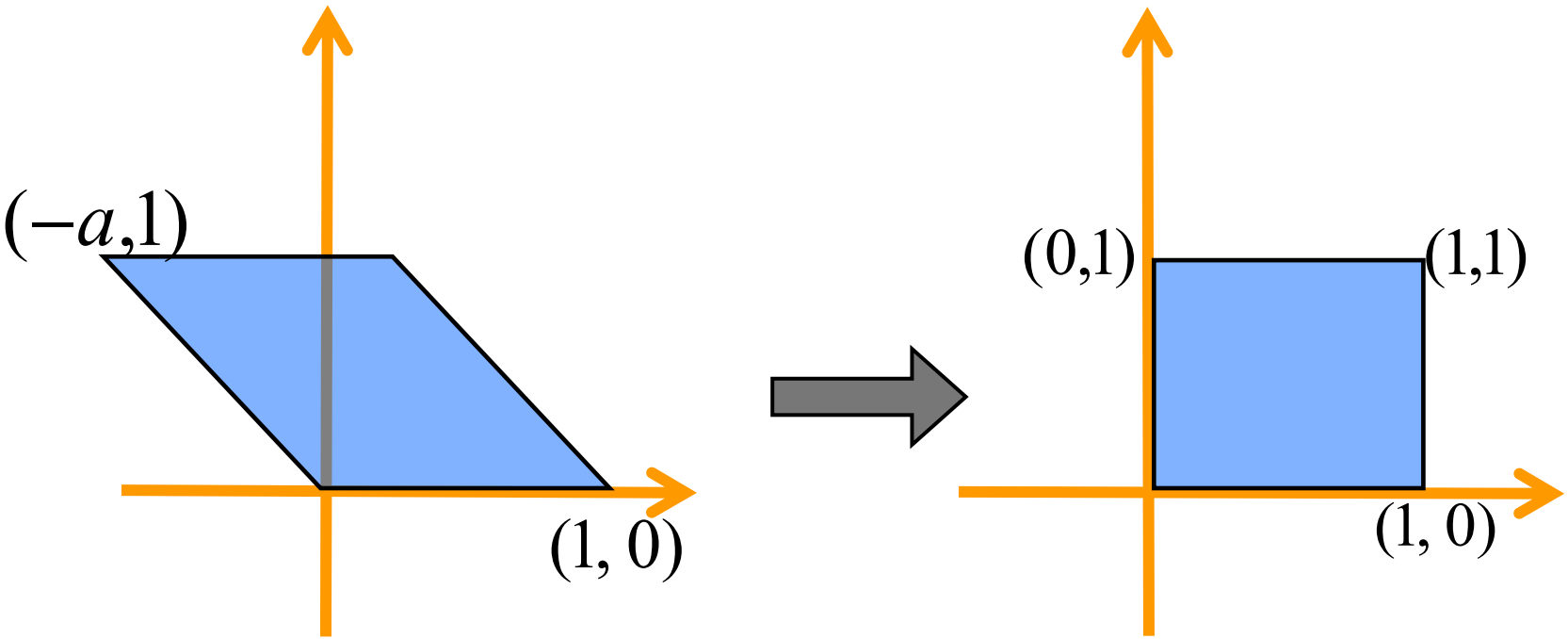
- Shear in x: keep the coordinate of y; change the coordinate of x by adding ay.

$$Sh_x = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ay \\ y \end{bmatrix}$$

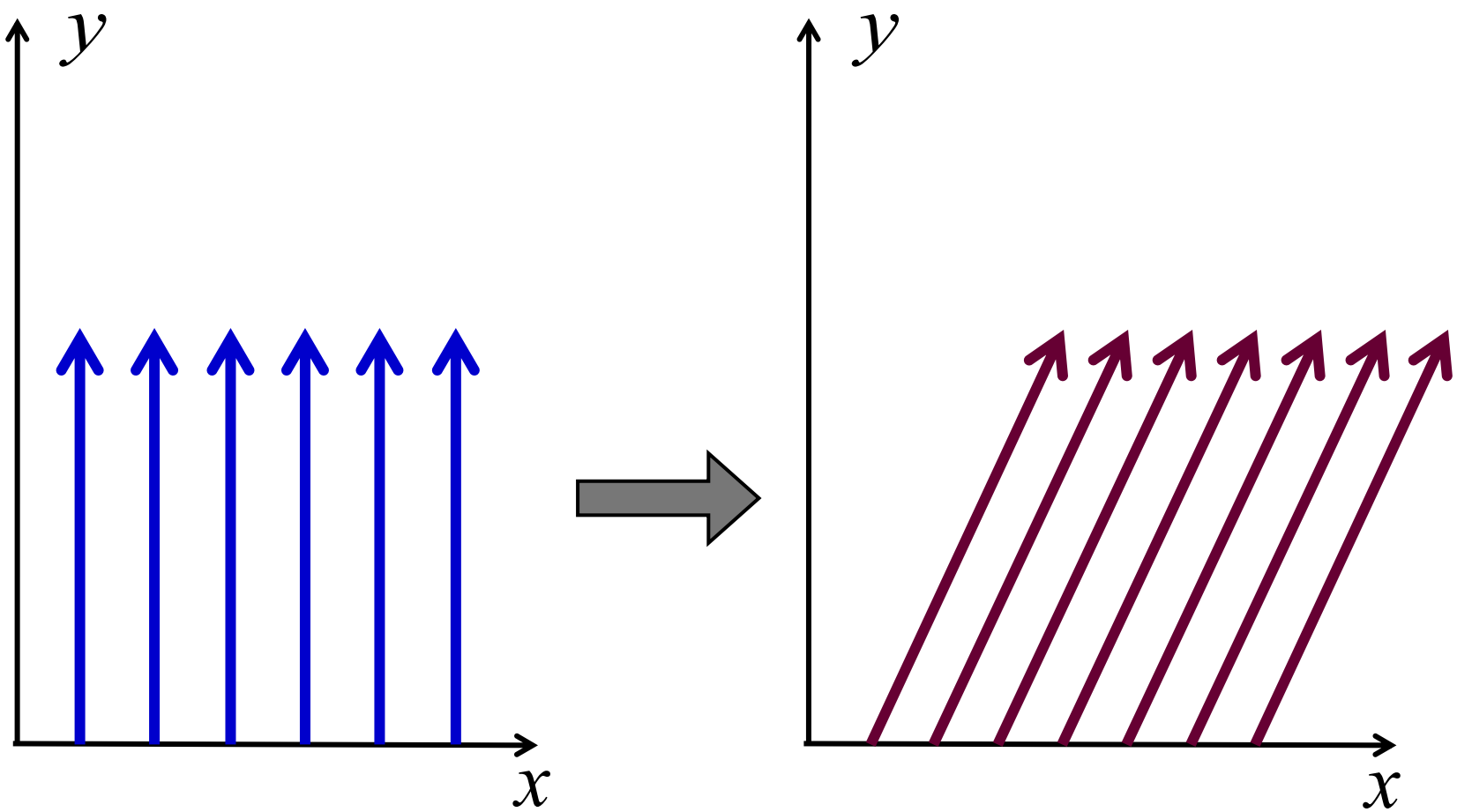
- Shear in y: keep the coordinate of x; change the coordinate of y by adding bx.

$$Sh_y = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ bx + y \end{bmatrix}$$

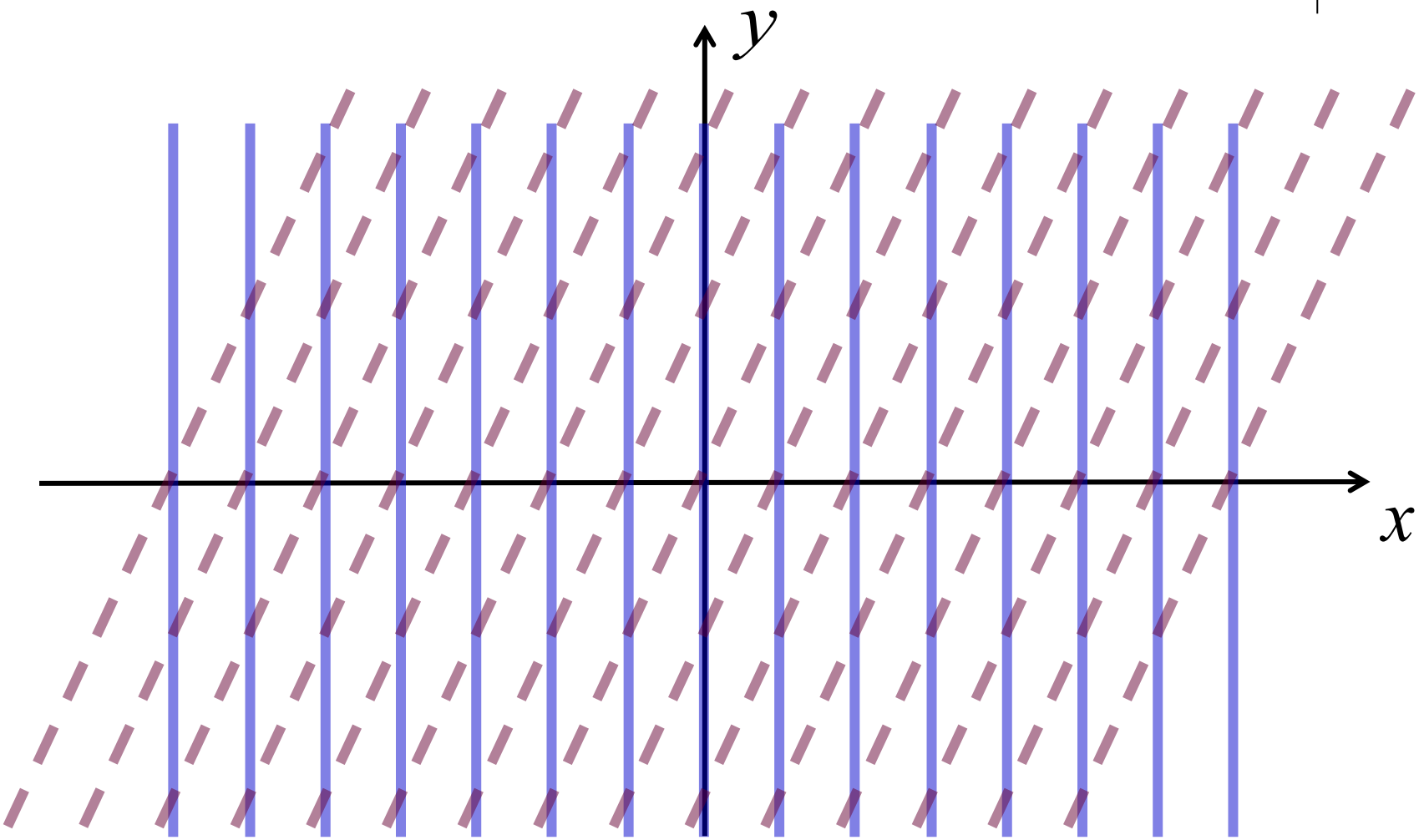
Geometric View of *Shear in x*



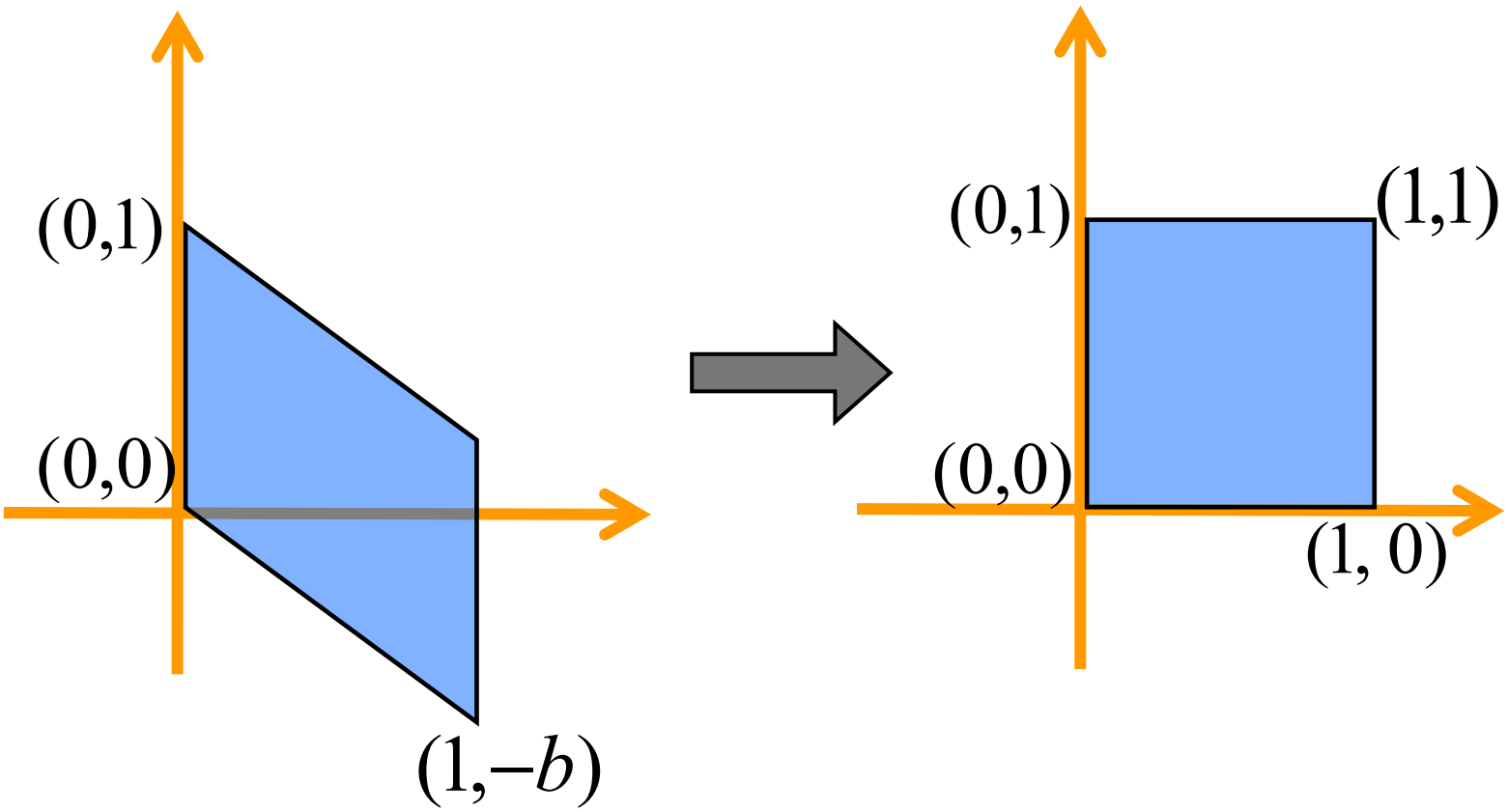
Another Geom View of *Shear in x*



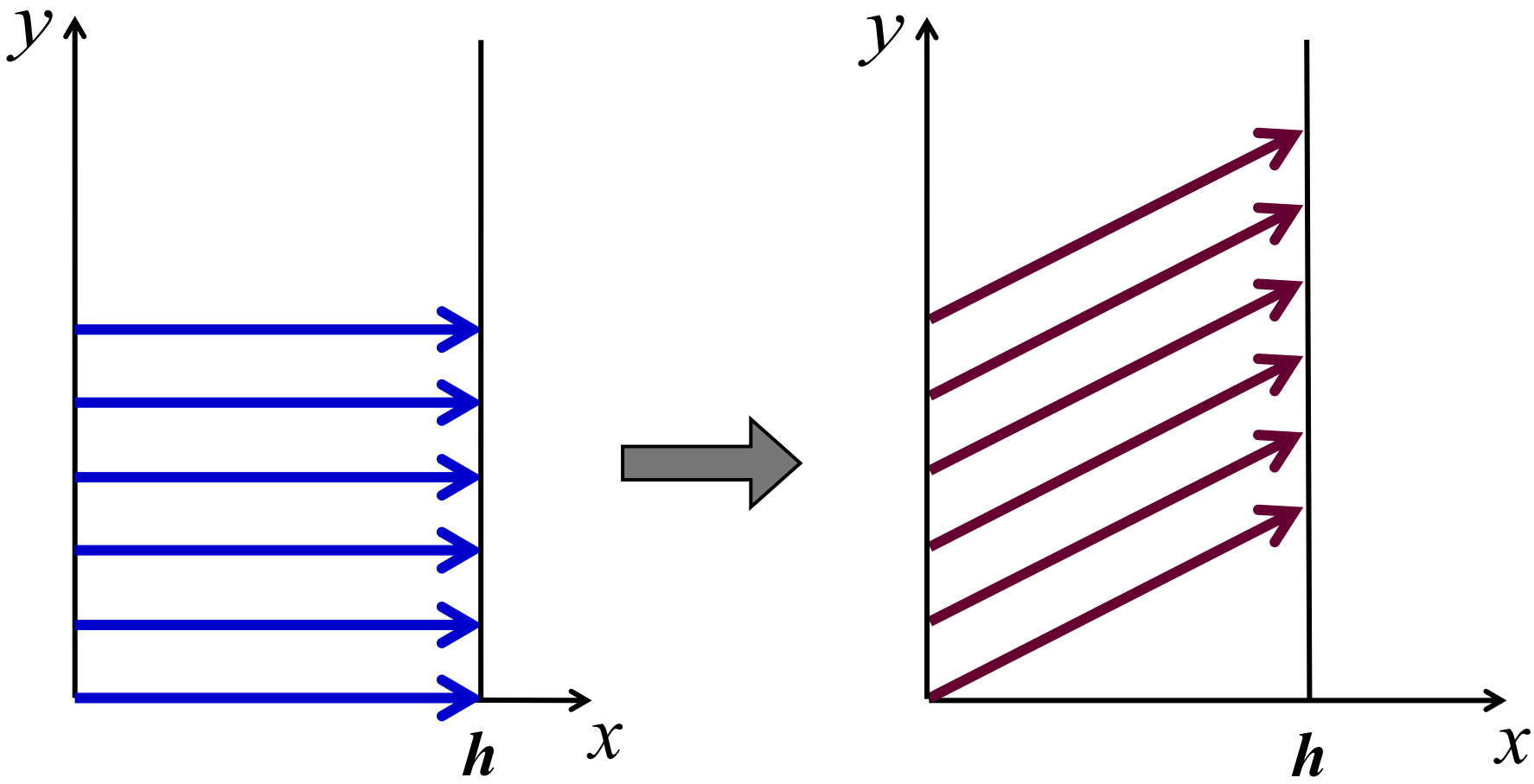
Another Geom View of *Shear in x*



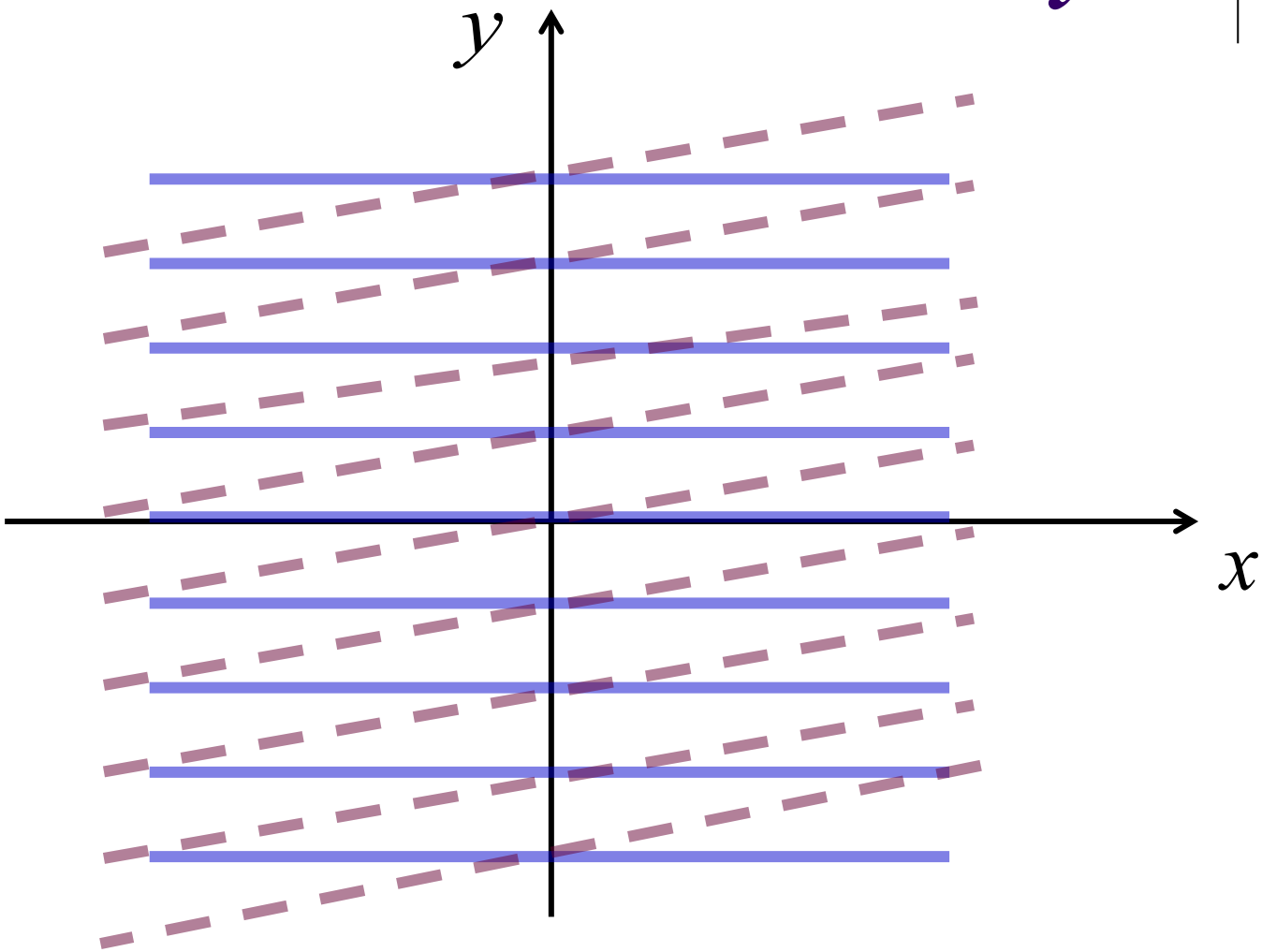
Geometric View of *Shear in y*



Another Geom View of *Shear in y*



Another Geom View of *Shear in y*



Introduce “Lazy 1”

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation in x

$$\begin{bmatrix} 1 & 0 & d_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + d_x \\ y \\ 1 \end{bmatrix}$$

Translation in y

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & d_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y + d_y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous Coordinates

$$\begin{bmatrix} \lambda x \\ \lambda y \\ \lambda \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \leftrightarrow \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{for } \lambda \neq 0$$

Homogeneous Coordinates

For $\lambda \neq 0$,

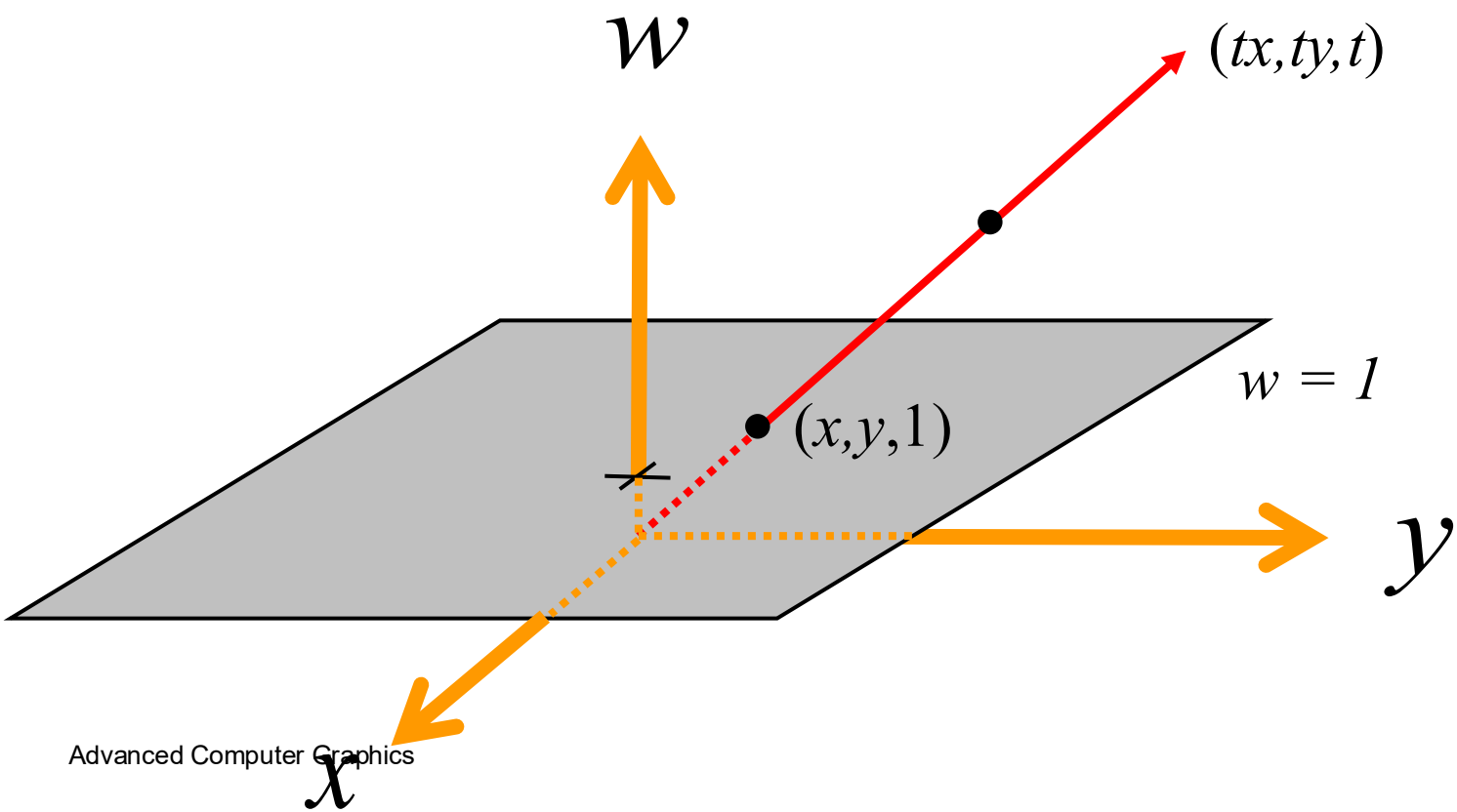
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1/\lambda \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \\ 1 \end{bmatrix} \Leftrightarrow \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$$

Homogeneous term affects overall scaling

Homogeneous Coordinates

An infinite number of points correspond to $(x,y,1)$.

They constitute the whole line (tx,ty,t).



Using Homogeneous Coord's

- *Shear* in $3D$
- Effects *translation* in $2D$
- We have used a *linear transformation* (*shear*) in $3D$ to effect a *nonlinear transformation* (*translation*) in $2D$

Translation by $\vec{d} : T(\vec{x}) = \vec{x} + \vec{d}$

$$T(\vec{u} + \vec{v}) = (\vec{u} + \vec{v}) + \vec{d}$$

$$T(\vec{u}) + T(\vec{v}) = (\vec{u} + \vec{d}) + (\vec{v} + \vec{d})$$

$$= \vec{u} + \vec{v} + 2\vec{d}$$

$$= T(\vec{u} + \vec{v}) + \vec{d}$$

$$\boxed{T(\vec{u} + \vec{v}) \neq T(\vec{u}) + T(\vec{v})}$$

Elementary Transformations

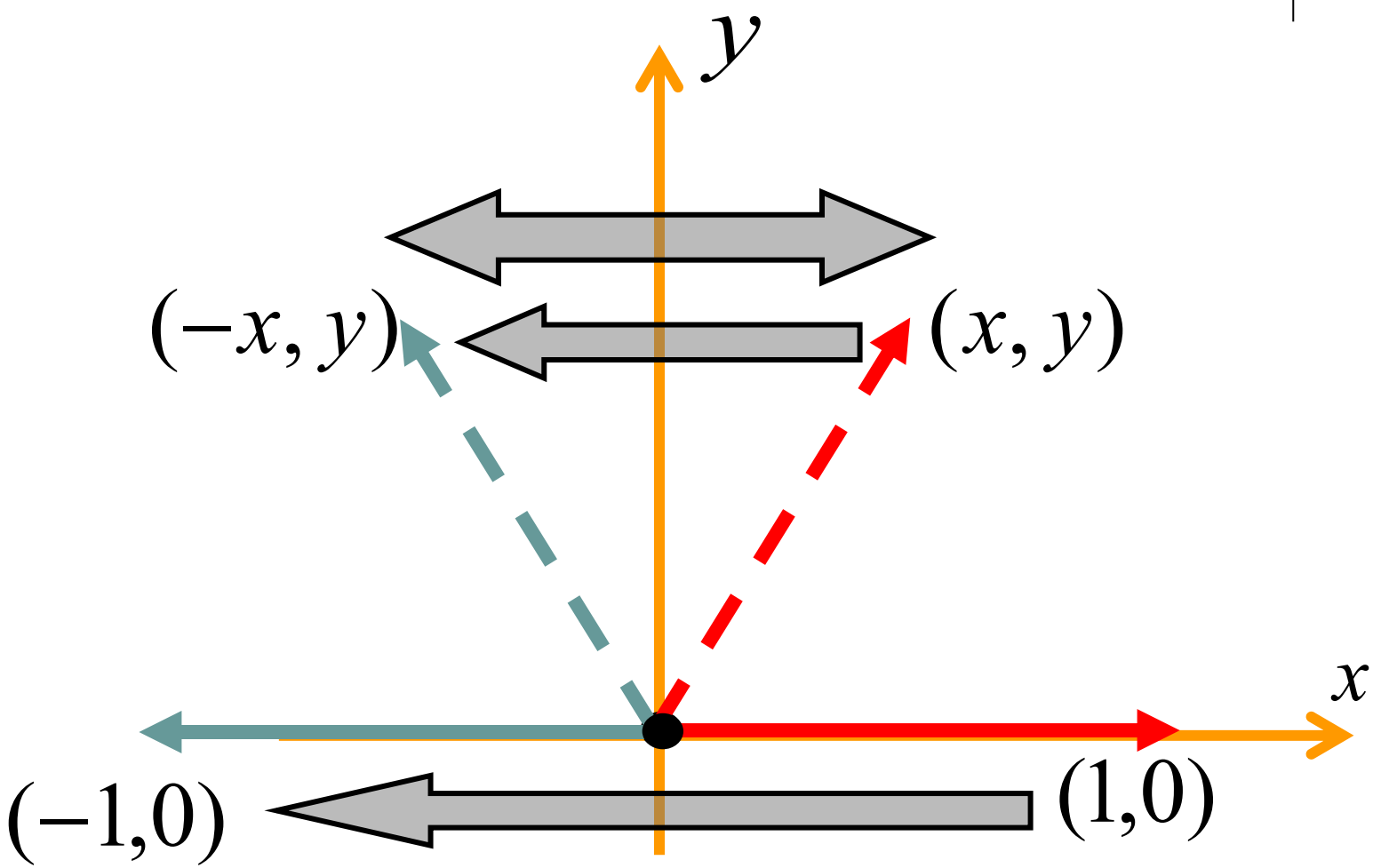
- *Scale*: $S_{\lambda_x}(v),\ S_{\lambda_y}(v)$
- *Rotate*: $R_{\theta_x}(v),\ R_{\theta_y}(v)$
- *Translate*: $T_{d_x}(v),\ T_{d_y}(v)$
- *Shear*: $Sh_{\lambda_x}(v),\ Sh_{\lambda_y}(v)$
- *Reflect*: $Rf_x(v),\ Rf_y(v)$

Refection about *y-axis*

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\boxed{x \leftrightarrow -x}$$

Reflection about *y-axis*

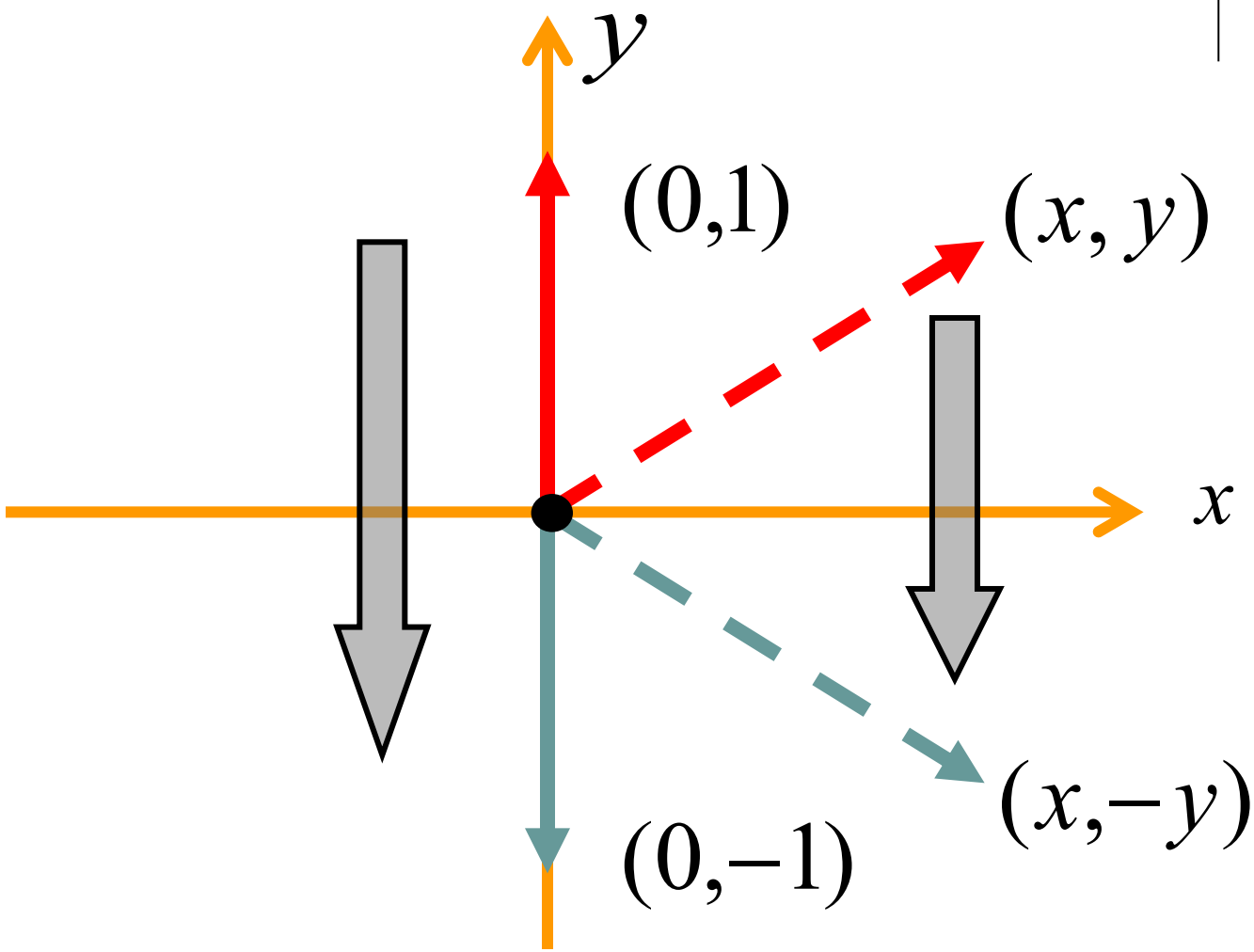


Reflection about *x-axis*

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\boxed{y \leftrightarrow -y}$$

Reflection about *x-axis*

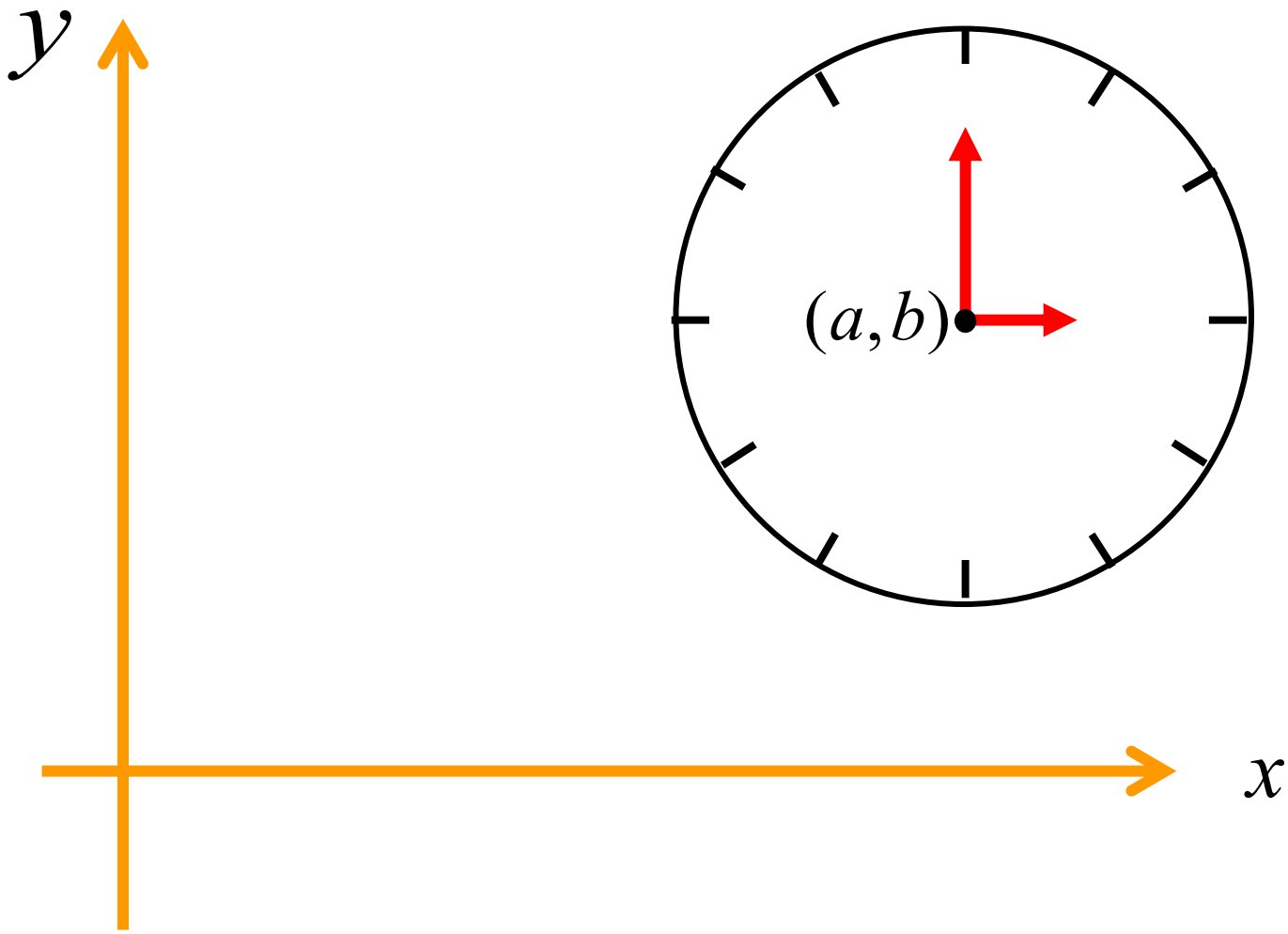


Is *Reflection* "*Elementary*"?

- Can we effect reflection in an elementary way?
- (More elementary means scale, shear, rotation, translation.)

Reflection = Scale *(-1)*

Ex: Advance clock hands



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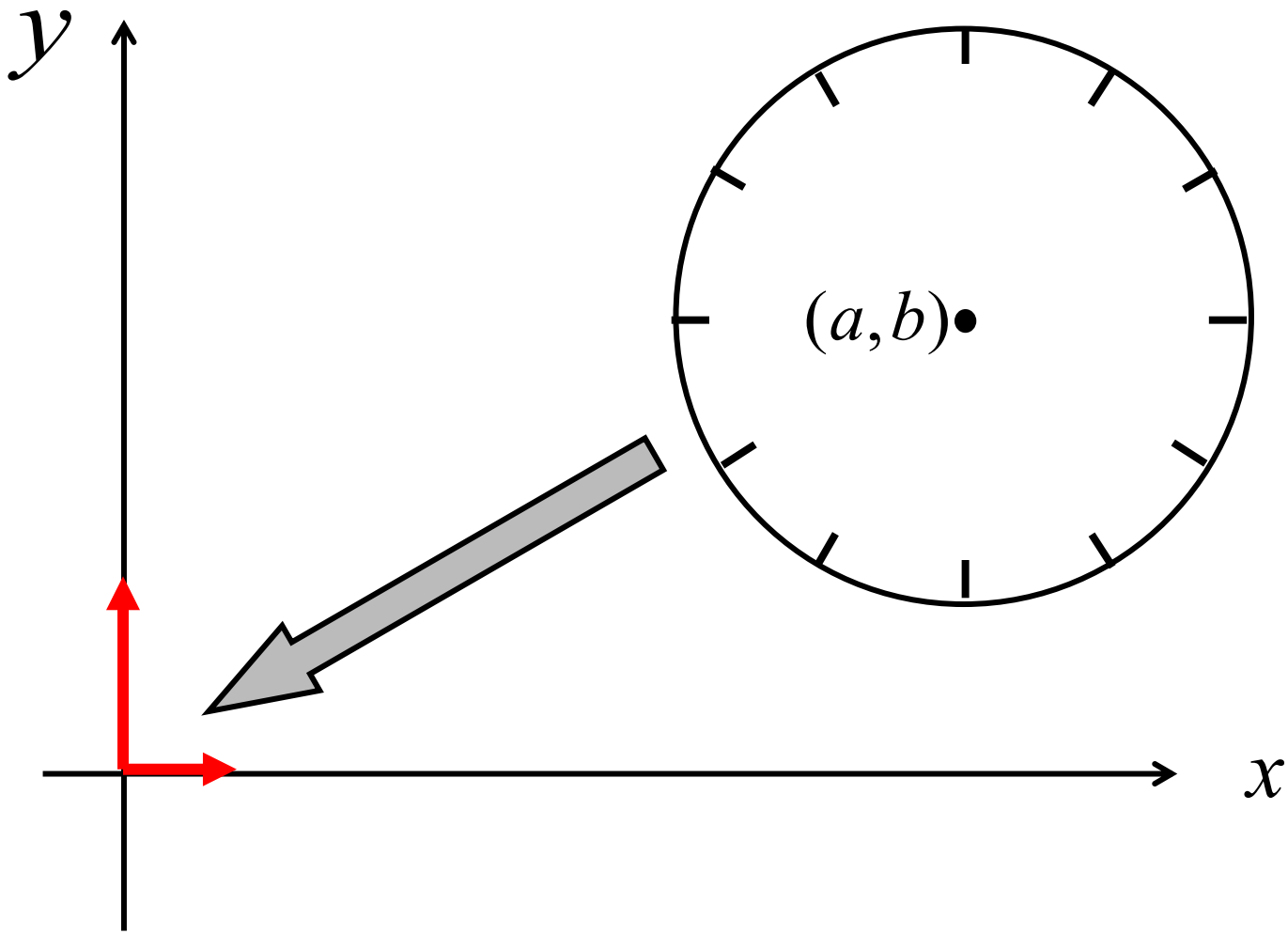

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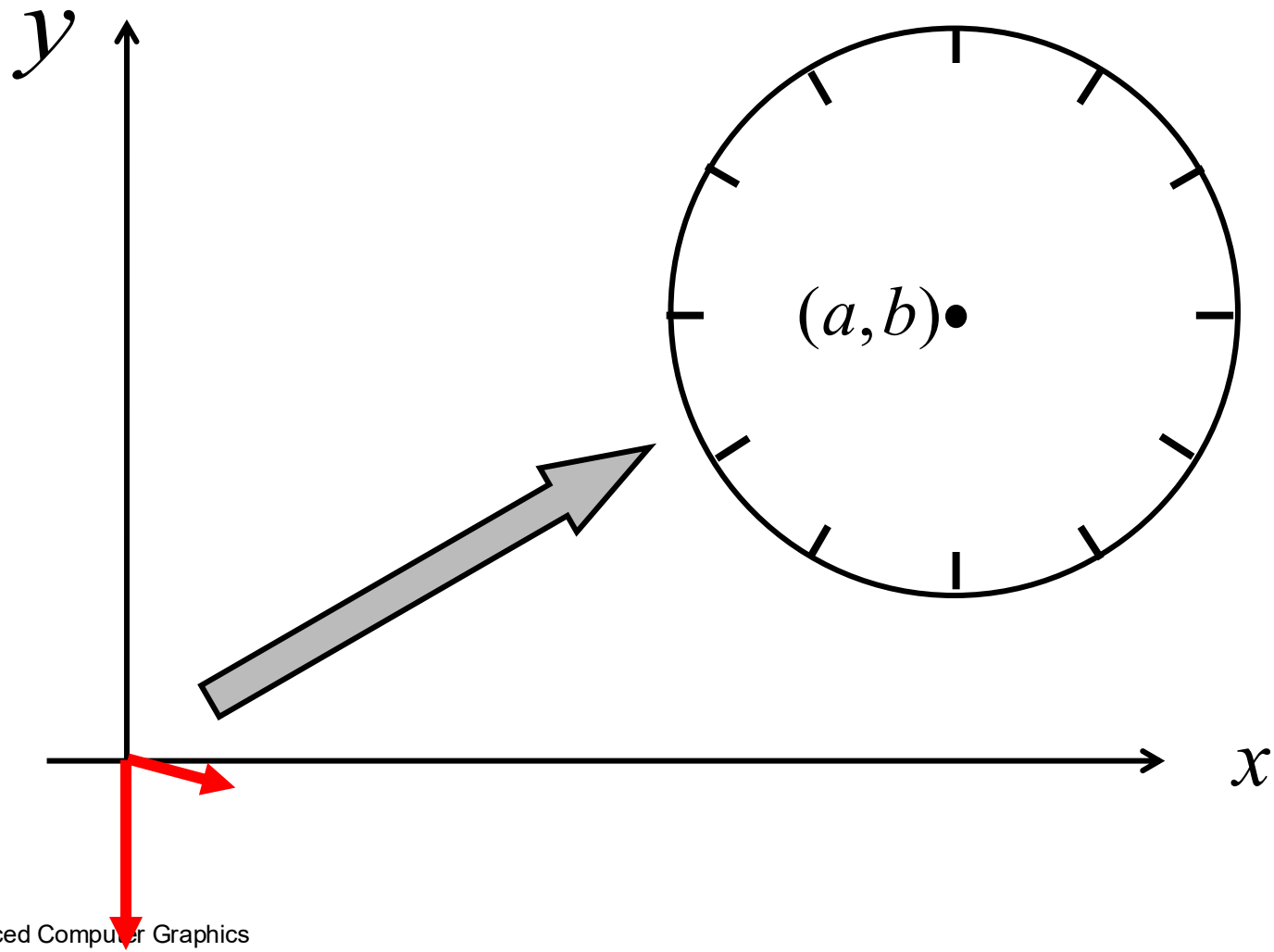

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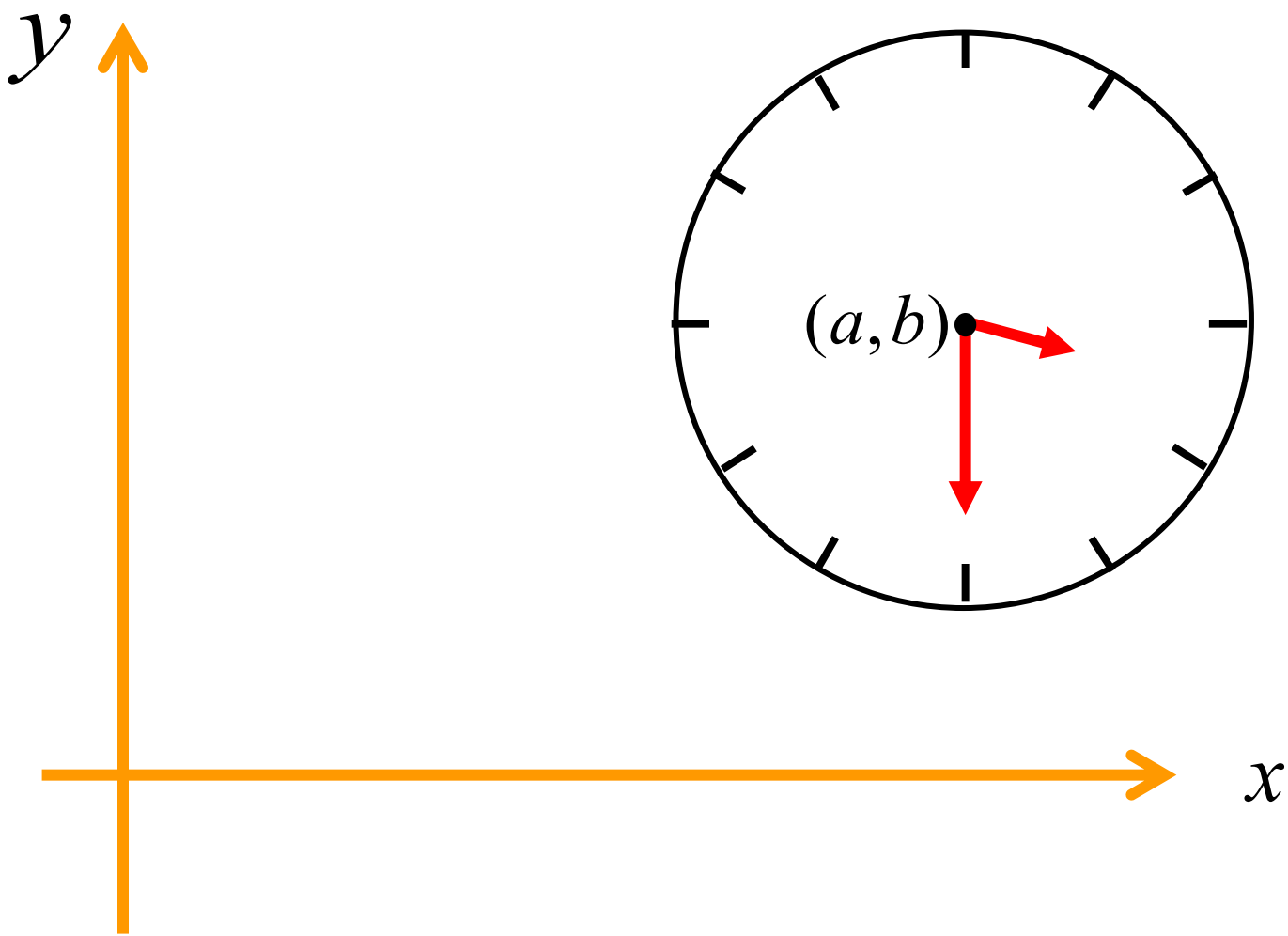

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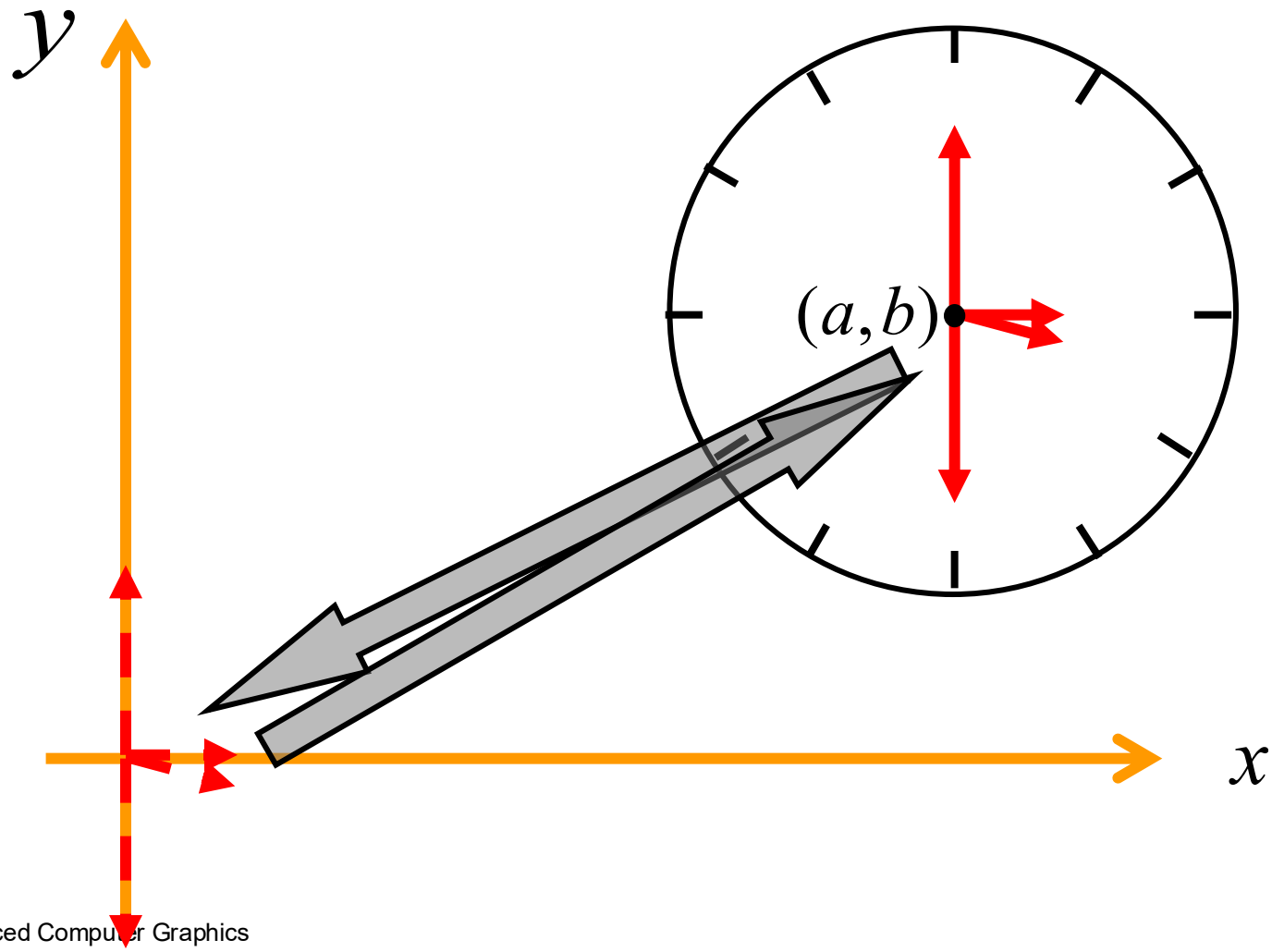

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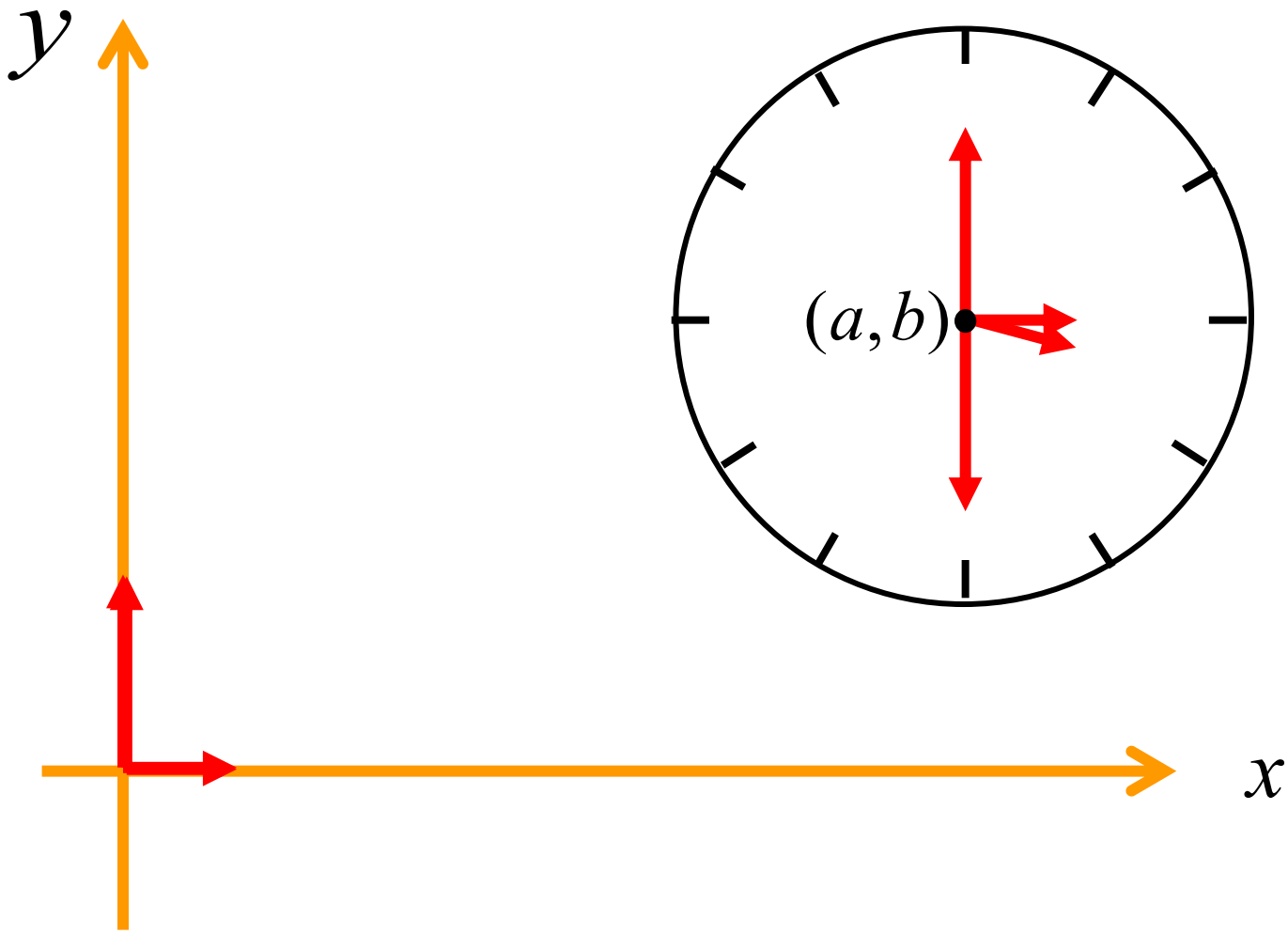


Clock *Transformations*

- *Translate* to *Origin*
- *Move* hand with *rotation*
- *Move* hand back to clock
- *Do* other hand

Clock Transformations

$$T_s = T(a,b)\, R(t)\, T(-a,-b)$$

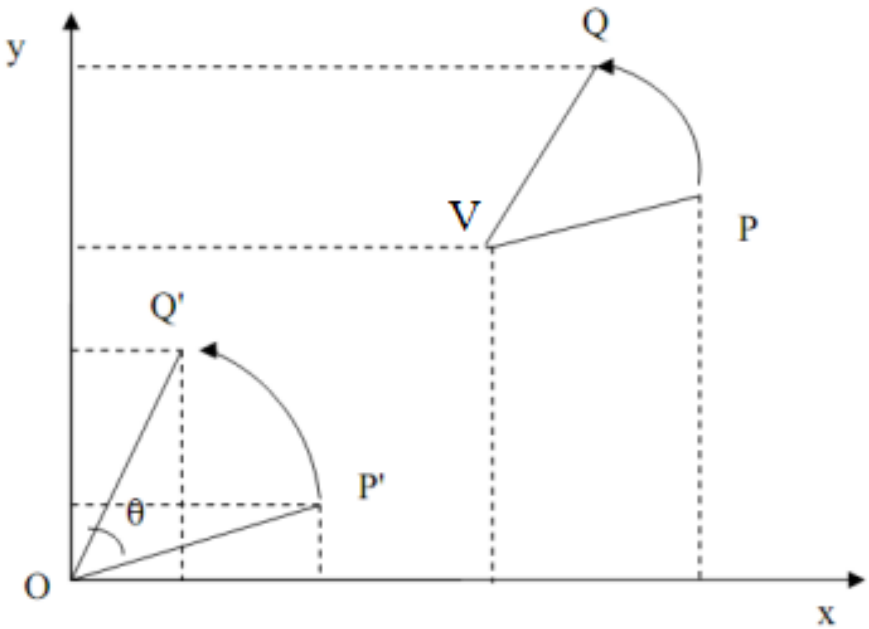
$$T_b = T(a,b)\, R(12 * t)\, T(-a,-b)$$

$$where \quad t = -15^\circ$$

Clock Transformations

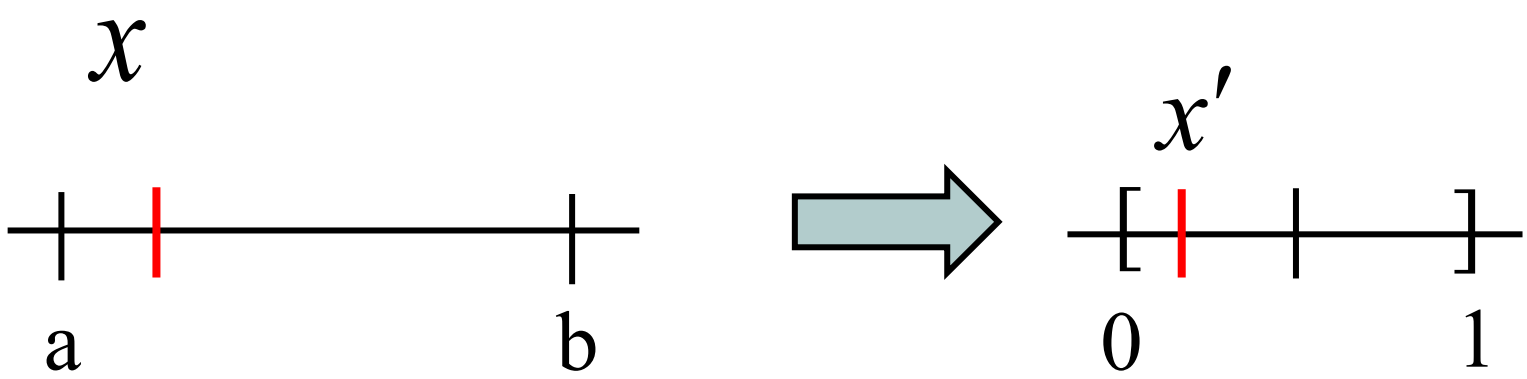
$$\underbrace{\begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Translate Back}} \underbrace{\begin{bmatrix} \cos(t) & -\sin(t) & 0 \\ \sin(t) & \cos(t) & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Rotate About Origin}} \underbrace{\begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Translate to Origin}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

example



- P(P.x, P.y)
- Rotate around V an angle θ to Q(Q.x, Q.y)
- Determine Q:
  - T(-V.x, -V.y) from P to P':
    P' = P+V or P'.x = P.x-V.x;
    P'.y = P.y-V.y
  - $R_\theta(O)$ from P' to Q':
    Q'.x = P'.x.cosθ - P'.y.sinθ
    Q'.y = P'.x.sinθ + P'.y.cosθ
  - T(V.x,V.y) from Q' to Q:
    Q.x = Q'.x + V.x
    Q.y = Q'.y + V.y

Map: $[a,b] \Rightarrow [0,1]$



Map: $[a,b] \Rightarrow [0,1]$

- Translate to Origin

$$[a,b] \rightarrow [a-a, b-a] = [0, b-a]$$

- Map x to translated interval

$$x \rightarrow x - a$$

Map: $[a,b] \Rightarrow [0,1]$

$$\underbrace{\begin{bmatrix} \frac{1}{b-a} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{S_x\left(\frac{1}{b-a}\right)} \underbrace{\begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{T_x(-a)} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \left(\frac{x-a}{b-a}\right) \\ y \\ 1 \end{bmatrix}$$

Map: $[a,b] \Rightarrow [0,1]$

- Normalize the interval

$$[0, b-a] \rightarrow \frac{1}{b-a}[a-a, b-a] = [0,1]$$

- Map x to normalized interval

$$x \rightarrow \frac{x-a}{b-a}$$

Just Look at ?[1]

$$\underbrace{\begin{bmatrix} \left(\dfrac{1}{b-a}\right) & 0 \\ 0 & 1 \end{bmatrix}}_{S_x\left(\frac{1}{b-a}\right)} \underbrace{\begin{bmatrix} 1 & -a \\ 0 & 1 \end{bmatrix}}_{T_x(-a)} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} \left(\dfrac{x-a}{b-a}\right) \\ 1 \end{bmatrix}$$

This is a homogeneous form for 1D

Recall from before…

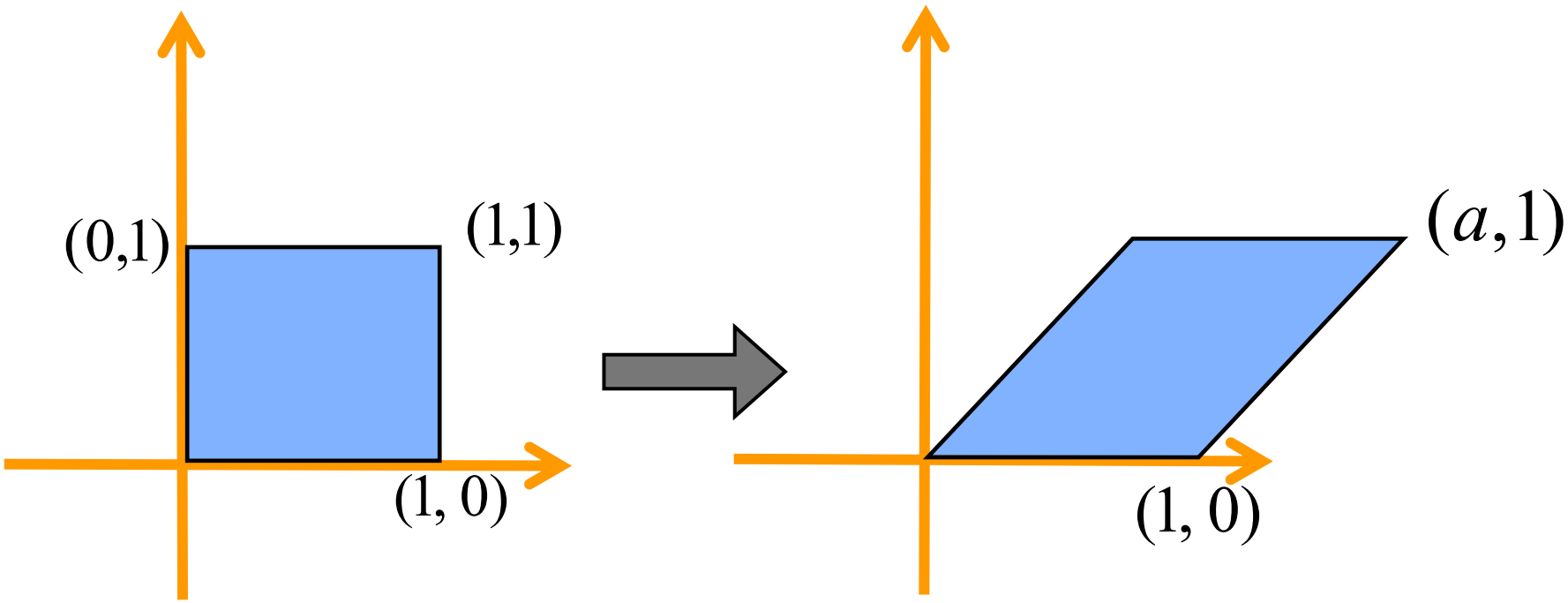
Shear in x:

$$Sh_x = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ay \\ y \end{bmatrix}$$

Shear in y:

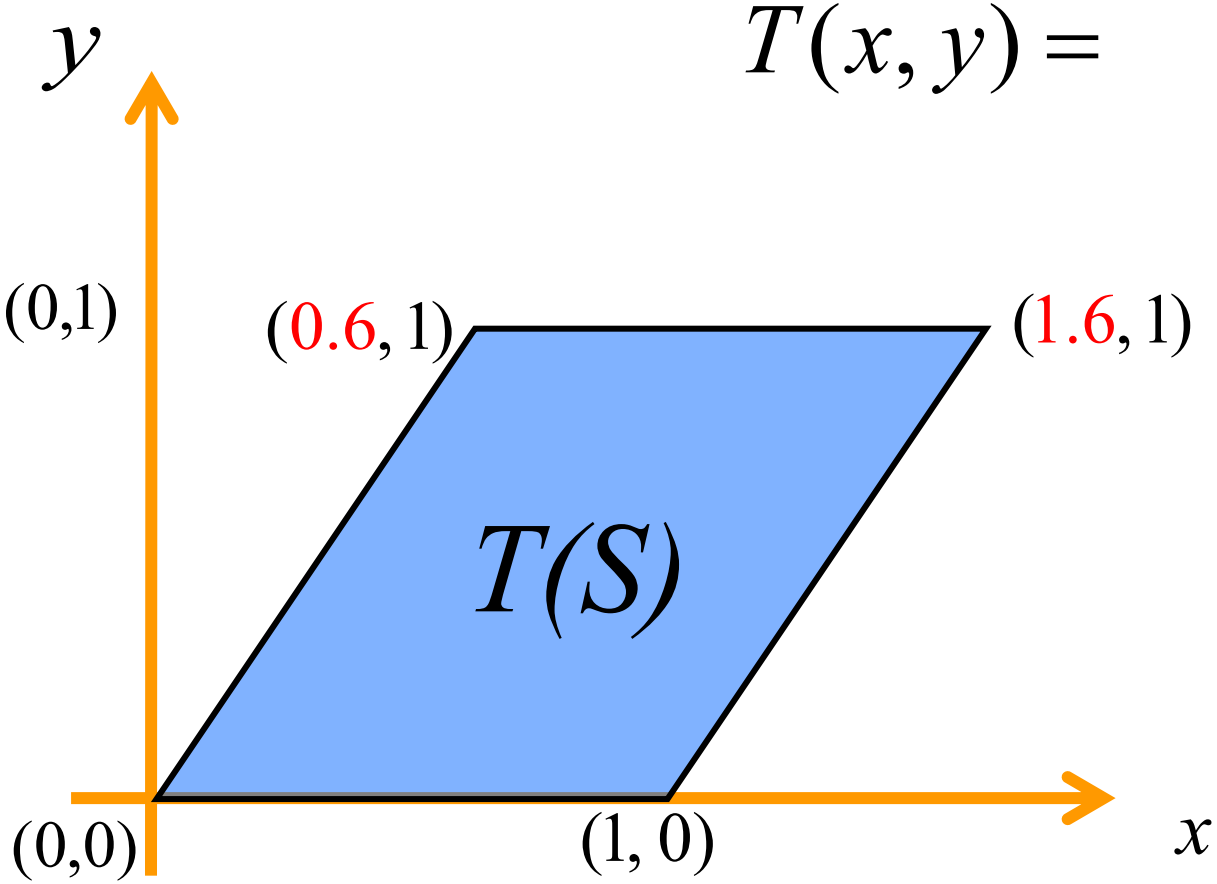
$$Sh_y = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ bx + y \end{bmatrix}$$

Geometric View of *Shear in x*

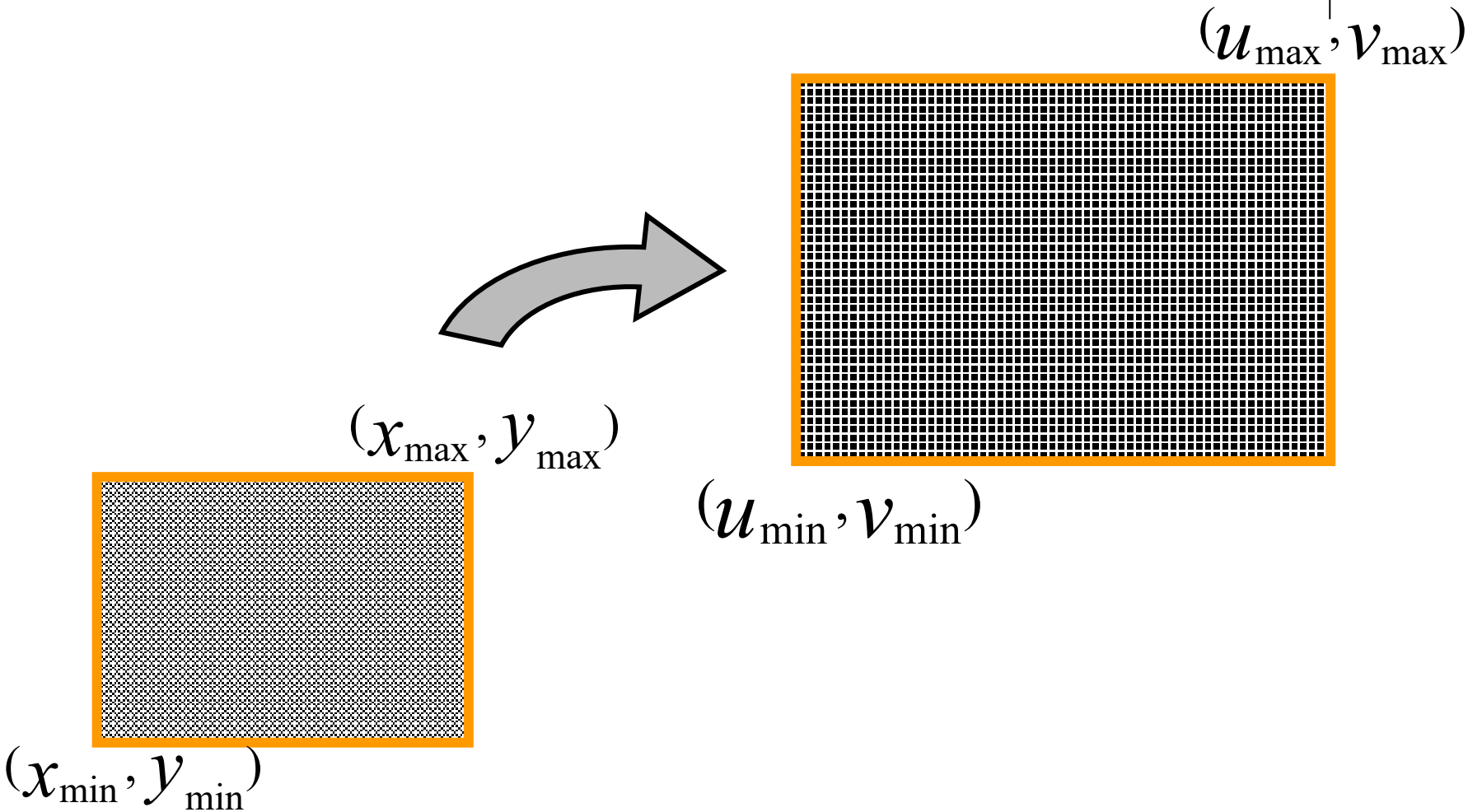


Consider

$$T(x, y) = \begin{bmatrix} x + 0.6y \\ y \end{bmatrix}$$



Now Map Rectangles

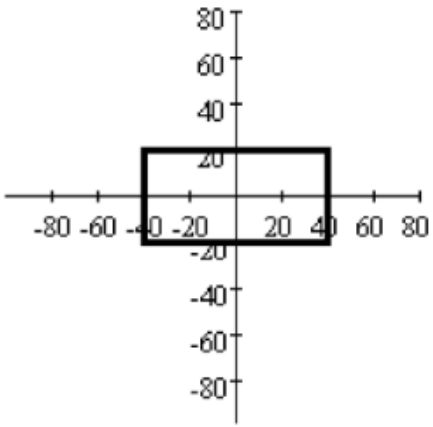


Transformation in x and y

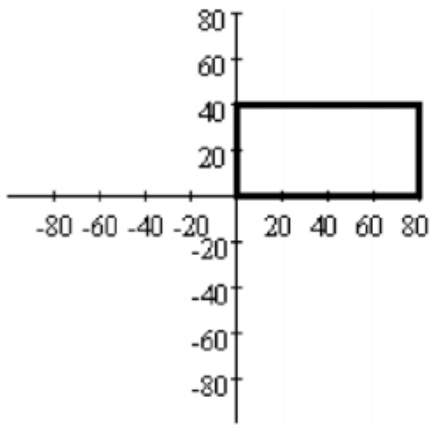
$$\begin{bmatrix} 1 & 0 & u_{\min} \\ 0 & 1 & v_{\min} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_x & 0 & 0 \\ 0 & \lambda_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x_{\min} \\ 0 & 1 & -y_{\min} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

where, $\lambda_x = \left(\dfrac{u_{\max} - u_{\min}}{x_{\max} - x_{\min}} \right)$, $\lambda_y = \dfrac{v_{\max} - v_{\min}}{y_{\max} - y_{\min}}$

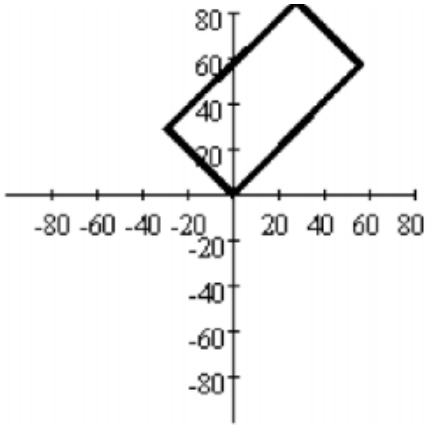
Examples:



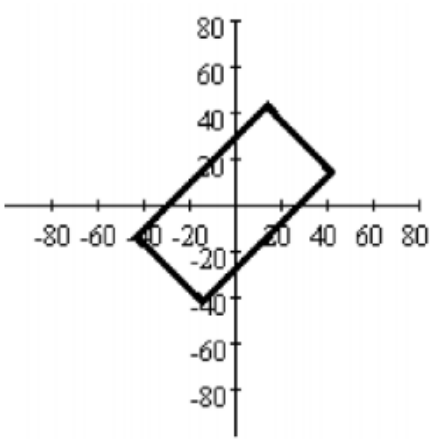
Original



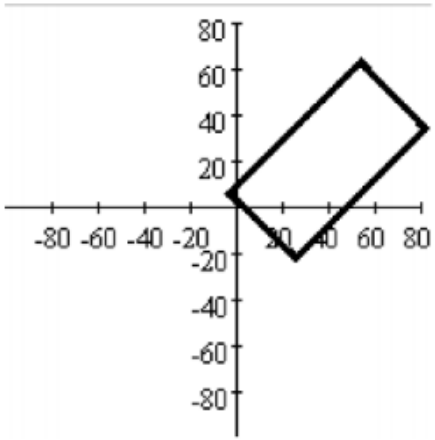
Translation



1. Translation, 2. Rotation



Rotation



1. Rotation, 2. Translation

This is *Viewport Transformation*

- Good for mapping objects from one coordinate system to another
- This is what we do with windows and viewports

3D Transformations

- *Scale* $S_x(\lambda),\ S_y(\lambda),\ S_z(\lambda)$
- *Rotate* $R_x(\theta),\ R_y(\theta),\ R_z(\theta)$
- *Translate* $T_x(d), T_y(d), T_z(d)$
- *Shear* $Sh_x(d),\ Sh_y(d),\ Sh_z(d)$

3D Scale in x

$$S_x(\lambda) = \begin{bmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Scale in x

$$S_x = \begin{bmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \lambda x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Scale in y

$$S_y(\lambda) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ \lambda y \\ z \\ 1 \end{bmatrix}$$

3D Scale in z

$$S_z(\lambda) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda z \\ 1 \end{bmatrix}$$

Overall 3*D Scale*

$$S(\lambda) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \left(1/\lambda\right) \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ \left(1/\lambda\right) \end{bmatrix}$$

Overall 3*D Scale*

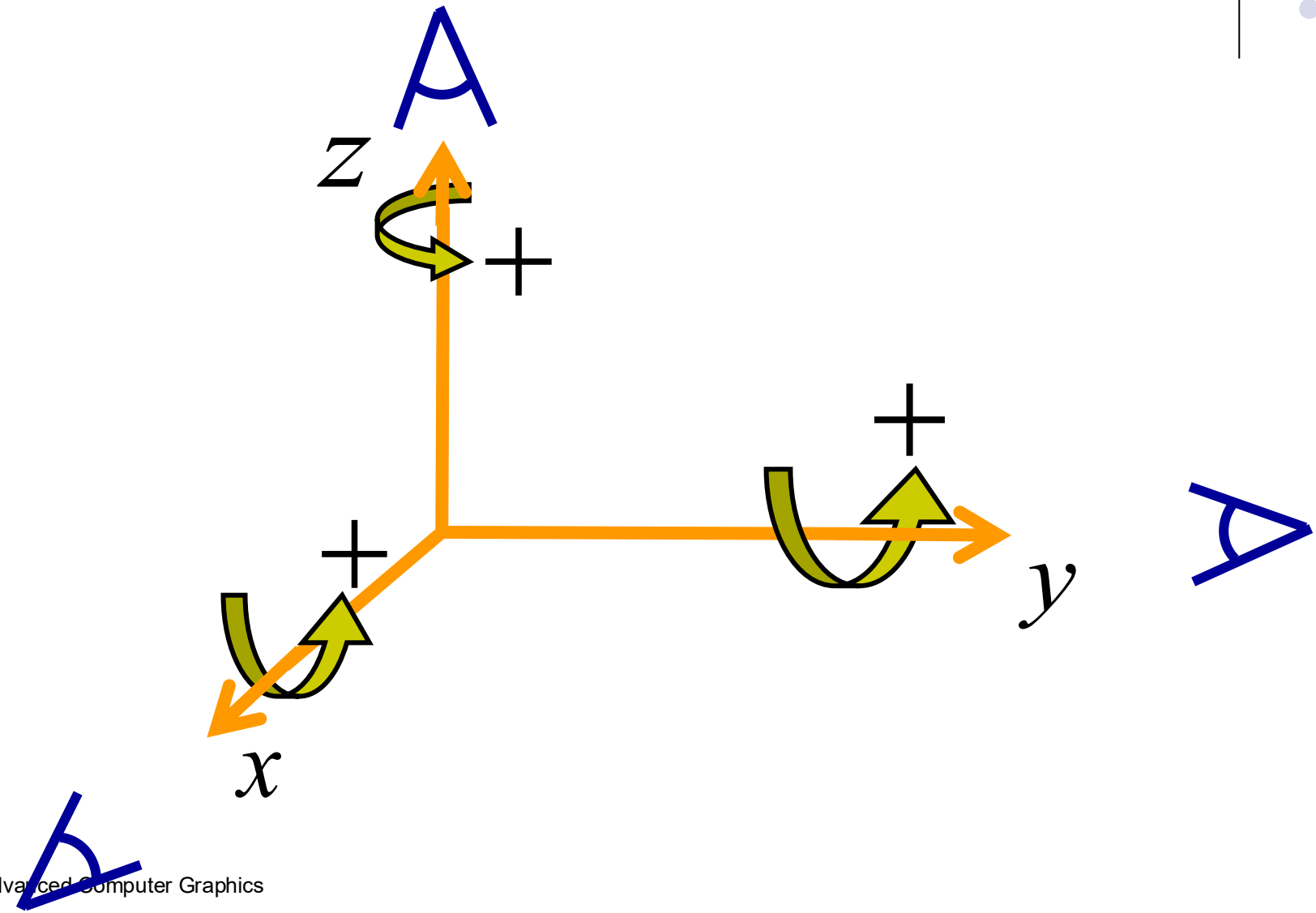
Same in x, y and z:

$$\begin{bmatrix} x \\ y \\ z \\ \left(1/\lambda\right) \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \end{bmatrix}$$

Positive *Rotation* in *3D*?

- Sit at $+\infty$ end of given axis
- Look at Origin
- CC Rotation is in *Positive* direction

3*D* Positive *Rotations*

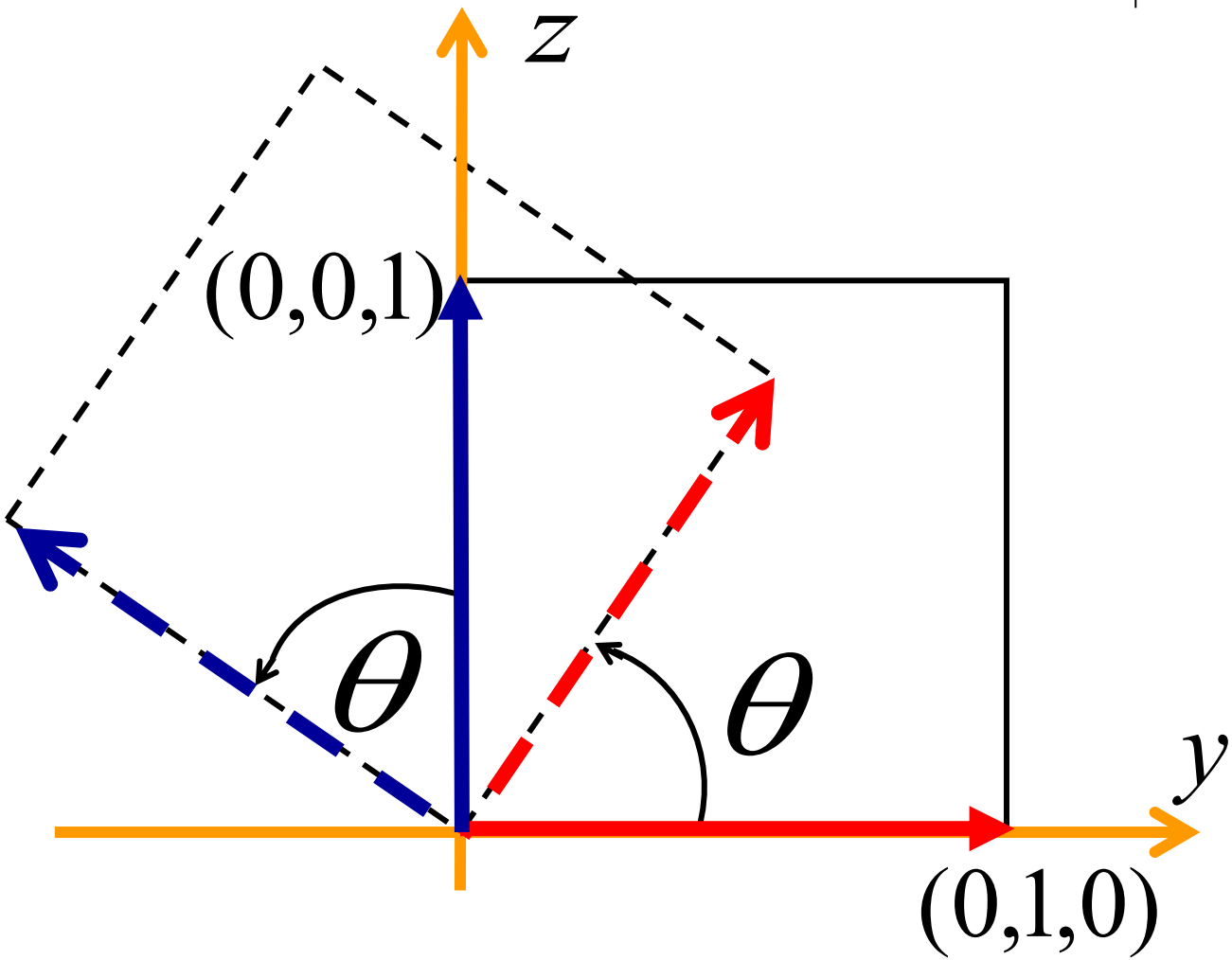


3D Rotation about *z-axis* by θ

We have already done this:

$$R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

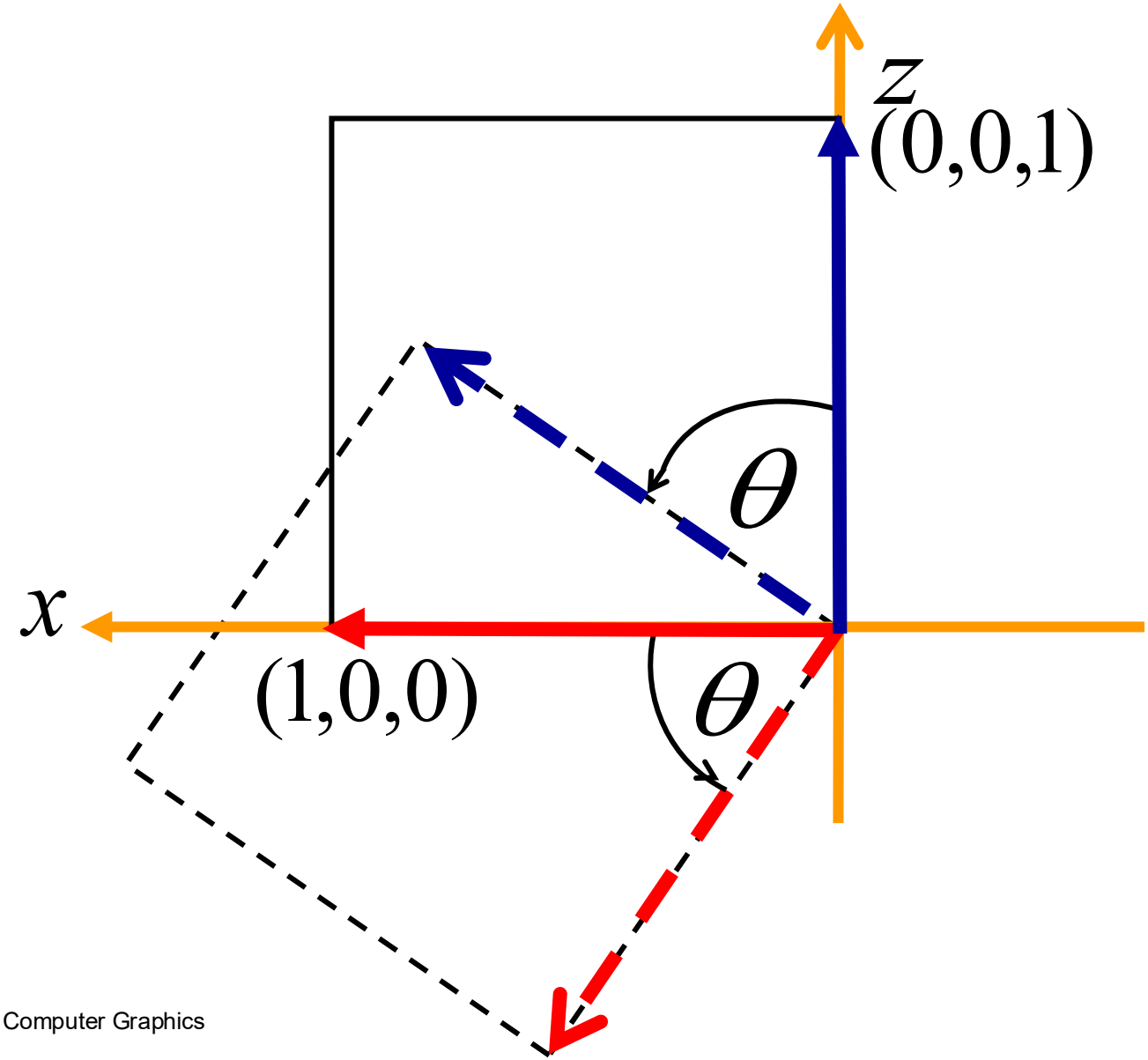
3*D Rotation* about *x-axis* by θ



3*D Rotation* about *x-axis* by θ

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3*D* *Rotation* about *y-axis* by θ



3D *Rotation* about *y-axis* by θ

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Elementary Transformations

- *Scale*: $S_{\lambda_x}(v),\ S_{\lambda_y}(v)$
- *Rotate*: $R_{\theta_x}(v),\ R_{\theta_y}(v)$
- *Translate*: $T_{d_x}(v),\ T_{d_y}(v)$
- *Shear*: $Sh_{\lambda_x}(v),\ Sh_{\lambda_y}(v)$
- *Reflect*: $Rf_x(v),\ Rf_y(v)$

Homework

Reading and revising the following chapters:

- Textbook: chapter 7
- Mathematics for CG: chapter 9
- Intro to CG using Java 2D & 3D: chapter 5